

SSC CGL (COMBINED GRADUATE LEVEL - INSPECTOR, ASO, TAX ASST) — SUBJECT COMPLETE QUESTION BANK

PRACTICE MATERIAL • VERIFIED QUESTION BANK

Total Questions: 250 • Mode: Single Subject Comprehensive Practice

- Q.1.** '©@'%@ 'YG 'éB')M'ò .y 20% 'Y@ "YC'iM'y? "TMK'%G '©0 •ù 'yC"TM#" "2 '©(" '©\$ 'éG•" "ù\$'%G '©M" \$"ù6'B " 'é@ 'Y0"
[English]: If the price of sugar increases by 20%, by what percent must a household reduce consumption so that expenditure remains unchanged?
A) 16.67% (16 !T%) B) 20%
C) 25% D) 15%
- Q.2.** •ù 'iA'Y>'%&"é0 'Y?'%@ "Y8"Ù\$" * " # R " R " &"2 "Ù0'é?'R ") (Successive Discounts) 'iG'I> "TMH-B 'Y2 "%.'IA")M'ò , 'ùM
[English]: A shopkeeper offers two successive discounts of 20% and 10%. What is the single equivalent discount?
A) 28% B) 30%
C) 25% D) 27%
- Q.3.** •ù 'ÉG•%."é("YM'ù>'©>" @ •Y*'%@ "Y8"Ù\$" • "2 "Ù0'ò .")2"Ù/ '©0 'ÉG'©("r "â &"é5"â " \$"â 9", * "IA 1 'Y?'yM" > 'YG "%M"
[English]: A dishonest dealer professes to sell at cost price but uses a false weight of 900g instead of 1kg. Find his profit percentage.
A) 11.11% (11 !Q%) B) 10%
C) 12.5% D) 9%
- Q.4.** 'YK•, "%0"é6"ò 8"é"é0'2 , "Ù/"é 'Y@ 'i0 "%G 8 "Y0"Ù7"1 'éG•" &"1 " (" 9"2 "é\$" 9"%d "Y9" "%0"é6"ò "ù\$'%G "Y0"Ù7"1 'éG•"
[English]: A sum doubles itself in 8 years at simple interest. In how many years will it become 4 times?
A) 24 "Y0"Ù7 / 24 Years B) 16 "Y0"Ù7 / 16 Years
C) 32 "Y0"Ù7 / 32 Years D) 20 "Y0"Ù7 / 20 Years
- Q.5.** 10,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R &" 8"r " 5" M"r "r 'YM" 5"9&"Ù"ò , "Ù/"é 'i0 "%>'y>" # 'ÉM'ù>'Â "â •)\$ "F-f'erence
[English]: Find the difference between CI and SI on 10,000 at 10% per annum for 2 years:
A) 100 B) 150
C) 1200 D) 150
- Q.6.** 'ù&"ò B = 2 : 3 'l%"â " C = 4 : 5 "TMK, 'IK A : B : C 'Y> •Y(" *é\$ 'YM'ù> "TMK'y>?
[English]: If A : B = 2 : 3 and B : C = 4 : 5, find A : B : C:
A) 8 : 12 : 15 B) 6 : 9 : 15
C) 8 : 10 : 15 D) 4 : 6 : 10
- Q.7.** A 'Y?'%@ 'Y>" M'ò "2 &"ù('éG•" \$'Y> B •TM8" "é0"Ù/ 'YK 15 'i?', .y '©B" > 'Y0 "%0 'i> "TMH-B &"1("1 'é?) " " , "é0"Ù/ 'YK 'Y"
[English]: A can do a piece of work in 10 days and B in 15 days. Working together, in how many days will they finish the work?
A) 6 'i?', ò b Fys B) 8 'i?', ò , Fys
C) 5 'i?', ò R Fys D) 7 'i?', ò r Fys
- Q.8.** '©>•* A •ù 'Y@ 'YK 6 '% 'ùG 'éG•" -" 8'Y\$â 9", " *é'¢ " "%G 8 '% 'ùG 'éG•" "é2" " 8'Y\$â 9"%d 'iK'%K•" *é'¢ 'R 8"é"
[English]: Pipe A fills a tank in 6 hours and Pipe B empties it in 8 hours. If both open together, in how many hours will the tank fill?
A) 24 '% 'ùG / 24 Hours B) 14 '% 'ùG / 14 Hours
C) 18 '% 'ùG / 18 Hours D) 12 '% 'ùG / 12 Hours
- Q.9.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'% 'ù> 'Y@ '©>" 8"r 'R •)-"r "2 "ù\$'%G "%G'Y ' .y '©>" " G'y@?
[English]: A 180m long train running at 54 km/h will cross a pole in how many seconds?
A) 12 "%G'Y ' ò " 6V6öæG0 B) 10 "%G'Y ' ò 6V6öæG0
C) 15 "%G'Y ' ò R 6V6öæG0 D) 18 "%G'Y ' ò , 6V6öæG0
- Q.10.** "i>•)\$ 'É2 'éG•" 'R ("é5 'Y@ '©>" "ù." ù •) "â 9", \$'Y> 'y>" > 'Y@ '©>" "ù." ù •) "â 9"%d 'y>" > 'YG •Y(" ")2 (Downstream)
[English]: Speed of a boat in still water is 10 km/h and stream speed is 2 km/h. How much time will it take to go 36 km downstream?
A) 3 '% 'ùG / 3 Hours B) 4 '% 'ùG / 4 Hours
C) 3.6 '% 'ùG / 3.6 Hours D) 4.5 '% 'ùG / 4.5 Hours
- Q.11.** '©M" %â S *"Ù0"é "9\$ "ù "% 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð
[English]: What is the average of first 50 natural numbers?
A) 25.5 B) 25
C) 26 D) 50

- Q.12.** '©?'I> 'I0 '©A'IM" " 5" M'I."é(•I/" "â '%A'©>'B R 2 "™H–B B 5" M"r , "é& •™('Y@ •I/" "â /"1 43 "Y0"Ù7 "™K'y>–B *""ù\$ "â " [English]: The ratio of present ages of father and son is 5 : 2. After 4 years, sum of their ages will be 43. Find father's present age:
- A) 25 "Y0"Ù7 / 25 Years B) 30 "Y0"Ù7 / 30 Years
C) 35 "Y0"Ù7 / 35 Years D) 20 "Y0"Ù7 / 20 Years
- Q.13.** 'ù&"ò 8•) "Ù/"â C f ³cs2 ("Â f" 8"r *)0"Ù#I "Y?"Ù>ÉM'ò 9"2Â \$"² v²r "r 8"Ù%"é('©0 'YL', 8"â •) •i 'y>? [English]: If number 481k673 is completely divisible by 9, find the digit in place of 'k':
- A) 7 B) 2
C) 5 D) 6
- Q.14.** 'yA'9('12 (7 y u - 3 u x) 'éG•" 'Y>•, "â •) (Unit Digit) 'YM'ù> "™K'y>? [English]: Find the unit digit in the expression (7 y u - 3 u x):
- A) 4 B) 6
C) 3 D) 7
- Q.15.** 'iK "%o 'iM'ù>'9 'Y>•Y(" ""é\$ 3 : 4 "™H 'I%"â "%o "â .–i8–b ,,,4b' B 9"%od •™('Y> ")f"%of (LCM) 'YM'ù> "™K'y>? [English]: The ratio of two numbers is 3 : 4 and their HCF is 4. What is their LCM?
- A) 48 B) 36
C) 24 D) 60
- Q.16.** ¹40 '©M" \$"ò "ù "Ù0"â 5"é2"r "é5"" "² "c *""Ù0'I? 'Y?'yM" > "Y>")G '©>"Y2 'YG "%o>'R "ù8 •Y(" ""é\$ 'éG•" ."ù2"é/"â "é 'Y? 'é [English]: In what ratio must rice at ¹40/kg be mixed with rice at ¹60/kg so that mixture costs ¹45/kg?
- A) 3 : 1 B) 1 : 3
C) 2 : 1 D) 3 : 2
- Q.17.** •ù "YC'IM'B " "" ?'y? 44 "%oG'é@ "™H–B "%o "â "Ù7"y\$"Ù0'12 (Area) 'YM'ù> "™K'y>? [English]: The circumference of a circle is 44 cm. What is its area?
- A) 154 "%oG'é@² / 154 cm² B) 308 "%oG'é@² / 308 cm²
C) 77 "%oG'é@² / 77 cm² D) 616 "%oG'é@² / 616 cm²
- Q.18.** •ù " 'A 5"9\$""Ù\$" / "i 'YA 'YG •i""é0 'Y@ 'IM" ?'ÉM'ù> 7 "%oG'é@ 'I%"â •) "é 24 "%oG'é@ "™H–B 6•) " "â 5'YM" *""97"Ù " / 'YM [English]: Base radius of a cone is 7 cm and height is 24 cm. Find its curved surface area:
- A) 550 "%oG'é@² / 550 cm² B) 500 "%oG'é@² / 500 cm²
C) 616 "%oG'é@² / 616 cm² D) 450 "%oG'é@² / 450 cm²
- Q.19.** A 'I0 B '%oG 'YM" . "i ¹20,000 'I0 ¹30,000 ") "é " 'R 5"Ù/"é*""é0 "iA" B 'Y?'ù>–B 5" M"r "r •)\$ 'éG•" " 2 ¹15,000 'Y>")>Ò 9" – [English]: A and B started a business investing ¹20,000 and ¹30,000 respectively. Total annual profit is ¹15,000. What is B's share?
- A) ¹9,000 B) ¹6,000
C) ¹7,500 D) ¹10,000
- Q.20.** 'ù&"ò , ² ÷, Ò B 9"2Â \$"² <" ² ÷<" "â ."é('YM'ù> "™K'y>? [English]: If $x + 1/x = 4$, find the value of $x^2 + 1/x^2$:
- A) 14 B) 16
C) 18 D) 12
- Q.21.** 'ù&"ò F â , = 4/3 "™K, 'IK sin ;, "â ."é('YM'ù> "™K'y>? [English]: If $\tan ;, \hat{O} B \hat{O} A \hat{v} \dagger B \text{---} 2 F \dagger R$ alue of $\sin ; f \hat{d}$
- A) 4/5 B) 3/5
C) 5/4 D) 3/4
- Q.22.** "YM'ù> 'É 24 ÷ 4 × (3 + 1) - 6 + 2 'Y> "%o0")\$'â ."é('YM'ù> "™K'y>? [English]: Simplify the expression $24 \div 4 \times (3 + 1) - 6 + 2$:
- A) 20 B) 18
C) 24 D) 16
- Q.23.** 'iK "YM'ù "Ù\$"ò 'R 9" 8"Ù%"é("%oG 'YM" . "i 4 'Y?'é@/'%o 'ù> 'I0 5 'Y?'é@/'%o 'ù> 'Y@ 'y\$"ò 8"r 5"ù*" @'B &"ù6"é •" . "y '©2'I0 [English]: Two persons start from the same point in opposite directions at 4 km/h and 5 km/h. After 3 hours, distance between them is:
- A) 27 'Y?'é@ / 27 km B) 15 'Y?'é@ / 15 km
C) 3 'Y?'é@ / 3 km D) 30 'Y?'é@ / 30 km

- Q.24.** •ù '©A'%>"R .y 'iK •™."Ù." &"Y>" %"yd 'É@'I("r 5"é2"r 'ém'é@'i5"é0 'YK 'YA"" 5"% 'é\$"'1 'Y> 60% 'é\$ 'é?")> 'I0 "Y9 1200 'é
[English]: In an election between two candidates, the winner got 60% of valid votes and won by 1200 votes. Total valid votes were:
A) 6,000 B) 5,000
C) 7,200 D) 4,800
- Q.25.** A 'I0 B 'Y?"%@ 'Y>" M'ò "2 "Ù0'é6•2 b &"ù('I0 8 'i?', .y 'Y0 "%@ 'IG "™H•)d '©B" G 'Y>" M'ò " 'É&")0" " C 9"%d A 'Y> "™H
[English]: A and B can do a work in 6 days and 8 days respectively. Total wages for the work is ₹1400. Find A's share:
A) ₹800 B) ₹600
C) ₹700 D) ₹750
- Q.26.** 'ù&"ò 'R 5"9\$"Ù\$ 'I0 •ù "Y0"Ù 'Y> '©0"ù."é* "%."é("™K, 'IK •™('YG 'YM"yG'IM" +")K•" "â '%A'©>'B "Ù/"â 9"1 "ãð
[English]: If the perimeter of a circle and a square are equal, what is the ratio of their areas?
A) 14 : 11 ("YC'IM'B "â, ' <"â' ò B 11 B) 11 : 14
C) 22 : 7 D) 1 : 1
- Q.27.** 'Y?"%@ "%@ 'iM'ù> 'YK 56 "%G 'Ù>'r &"y("r ** 6"y7"12 29 '©M" >'©M'B 9"1\$ "â 9"%d •™8" 8•) "Ù/"â "2, 8"r -"é 'iG'%G '©0 "iG
[English]: A number when divided by 56 gives a remainder of 29. What will be the remainder when divided by 8?
A) 5 B) 3
C) 7 D) 1
- Q.28.** ₹500 'Y@ 'y(" >"i? '©0 2 "Y0"Ù7 'Y> "%>'y>" # 'ÉM'ù>Â "C 9"%d "Y>" M"y?'R, "Ù/"é 'i0 'YM'ù> "™H?
[English]: The simple interest on ₹500 for 2 years is ₹40. What is the annual rate of interest?
A) 4% B) 5%
C) 6% D) 3.5%
- Q.29.** •ù 'É2"Ù2"y,"é '%G •Y*"%@ 11"Y@•" **é0" .y 100 " ('É("é 'É?"%8"r "%@ "â "%\$ 5 " ('É""Â 'ù>-B 5" '©>" @ 'YG 'É>'b
[English]: A batsman scores 100 runs in his 11th innings, increasing his average by 5 runs. Find his new average:
A) 50 " (/ 50 Runs B) 45 " (/ 45 Runs
C) 55 " (/ 55 Runs D) 60 " (/ 60 Runs
- Q.30.** 9 'I0 25 'Y> 'é""Ù/"é(" **é\$ " „ÖV â &÷ ÷ tional) 'YM'ù> "™K'y>?
[English]: What is the mean proportional between 9 and 25?
A) 15 B) 17
C) 12 D) 20
- Q.31.** A 'Y?"%@ 'Y>" M'ò "2 " &"ù("1 'éG•" " " "%@ 'Y>" M'ò "2 , &"ù("1 'éG•" **)0"â " 8'Y\$ "â 9"%d "YG 4 'i?'%K•" \$'R 'R 8"é0
[English]: A can do a work in 12 days and B in 18 days. They work together for 4 days, then A leaves. In how many days will B alone complete the remaining work?
A) 6 'i?'€ B) 8 'i?', f, F ys)
C) 10 'i?'€ D) 7 'i?'€
- Q.32.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'% "ù> 'Y@ 'y\$ "ò 8"r "" 0"™@ "™H-B 'R •)-r "2 **é0 'Y0'%G 'éG•" /"" "ù\$'%
[English]: A 180m long train is running at a speed of 54 km/h. How much time will it take to cross a pole?
A) 10 "%G'Y ' B) 12 "%G'Y ' f " 6V6ðæG2•
C) 15 "%G'Y ' D) 18 "%G'Y '
- Q.33.** ₹15,000 'Y@ 'y(" >"i? '©0 2 "Y0"Ù7 'YG ")?•ò R 5"é0"Ù7"ù 'i0 "%G '© "Ù0"YC'iM'y? 'ÉM'ù>Â " 8"é""é0'2, "Ù/"é 'Y> •Y 'I0 'Y
[English]: What is the difference between CI and SI on ₹15,000 for 2 years at 10% per annum?
A) ₹120 B) ₹150 (₹150)
C) ₹180 D) ₹200
- Q.34.** •ù 'iA'Y>%&"é0 'Y?"%@ "Y8"Ù\$" "r •) "ù\$ 'éB")M'ò * " # R " ") 'iG'%G 'YG 'É>'b -" # R "â 2"é- •Y0"Ù "ù\$ 'Y0'I> "™H-B /
[English]: A shopkeeper gives 20% discount on marked price and still gains 20%. If cost price is ₹500, what is the marked price?
A) ₹650 B) ₹700
C) ₹750 (₹750) D) ₹800
- Q.35.** •ù 'YH")G 'éG•" " Â S **%8"r " #R **%8"r "r 8"ù "Ù "r 2 4 : 5 'YG •Y(" **é\$ 'éG•" 9"% -B /'i? 'YH")G 'éG•" " 2 'y(" >"i? ₹170
[English]: A bag contains ₹1, 50p and 25p coins in the ratio 3 : 4 : 5. If total amount is ₹170, find the number of 50p coins:
A) 80 B) 100
C) 120 (120 coins) D) 150

- Q.36.** •ù ") 'Â 5"9\$"Ù\$" / 'ÉG") ('YG •i"é0 'Y@ 'IM" ?'ÉM'ù> 7 "%G'é@ 'I0 •© '©>•, 8"y." 9"%d •y8'Y> 'YA"" *97"Ù " / 'YM"yG'IM"
[English]: Find the Total Surface Area of a right circular cylinder with radius 7cm and height 10cm:
A) 648 "%G'é@² B) 748 "%G'é@² (748 cm²)
C) 848 "%G'é@² D) 924 "%G'é@²
- Q.37.** 'ù&"ò 6-â , + cos ;, Ò !£" 6÷2 , 'IK cos ;, Ò 6-â , 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: If $\sin ;, ^2 6\div 2 , = " 2 \cos ;, \hat{A} F\hat{+}V\hat{a} v\hat{t} B \longrightarrow F\hat{+}R \hat{a}lue \text{ of } \cos ;, \hat{O} 6-\hat{a} , ?$
A) " 2 sin ;£ B) " 2 tan ;£
C) sin ;£ D) 1
- Q.38.** 'iK "% 'iM'ù>'9 'Y> 'éf"%f (HCF) 15 'l%"â 2-i8-b „Ä4Ò" S 9"%d 'ù&"ò 'R 8•) "Ù/"â 3 9"2Â \$"2 &")8" @ "% 'iM'ù> 'YM'ù> "TMK
[English]: [HCF is 15, LCM is 150. If one number is 30, find the other number:
A) 45 B) 75
C) 60 D) 90
- Q.39.** "%? 'éM"%2"ù "ù\$ 'éG•" 8"r "É("%@ •ù •Y*" ?'éG'ò 8•) "Ù/"â „—' ational Number) "TMH?
[English]: [Which of the following is an irrational number?
A) " 4 B) " 9
C) " 7 D) 22/7
- Q.40.** '©>•, fÀ) 'Y?“, *"Ù0'Y>“ " 8•) "Ù/"â 9"fð
[English]: [What type of number is Pi (< "ð
A) '©0"ù."y/ "% 'iM'ù> B) •Y*" ?'éG'ò 8•) "Ù/"â „—' ational Number)
C) '©B" M'9>) D) '©M" >'YC'B 8•) "Ù/"â
- Q.41.** 'iM"Y?'%>'B 8'é@'Y0'2 <" 2', 2 2 Ò "r .")2 "Y>"%M'I5"ù 'I0 "%o. "é(("TMK•) "r /i?:
[English]: [Roots of $ax^2 + bx + c = 0$ are real and equal if:
A) $b^2 - 4ac > 0$ B) $b^2 - 4ac = 0$
C) $b^2 - 4ac < 0$ D) $b^2 - 4ac = 1$
- Q.42.** 'ù&"ò &"Ù5"ù "é\$ "%o." " # $2x^2 - 4x + k = 0$ 'YG 'éB"" , " >'É0 "TMK•"Â \$"2 2 "â . "é('YM'ù> "TMK'y>?
[English]: [If roots of $2x^2 - 4x + k = 0$ are equal, find value of k:
A) 1 B) 2
C) 4 D) -2
- Q.43.** 'iM"Y?'%>'B , "TMA'©& $x^2 - 3$ 'YG "iB'%M'ù (Zeroes) 'YM'ù> "TMK•) "sð
[English]: [Find zeroes of polynomial $x^2 - 3$:
A) 3, -3 B) + " 3, - " 3
C) " 3, " 3 D) 9, -9
- Q.44.** 'ù&"ò &"Ù5"ù "é\$ 'É9" *b <" 2', 2 2 "r 6")("Ù/'R ± 'I0 ; " 9"1 , 'IK (; 2 2) 'I0 ; 2 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: [Sum and product of zeroes ; æB 2 of $ax^2 + bx + c$:
A) -b/a 'l%"â 2ð B) b/a 'l%"â Ò2ð
C) c/a 'l%"â Ò"ð D) -c/a 'l%"â "ð
- Q.45.** " H'i?'R 8'é@'Y0'2 /" "Ù. a •x + b •y + c • = 0 'l%"â ' , 2 " " 2 2 " Ò "â 'iM"Y?'l@'ò 9"" ...Væ— VR 6öÇWF—öâ' 'Â 9"1\$ "â 9" fð
[English]: [Condition for a unique solution of pair of linear equations:
A) $a \cdot a , " b \cdot b ,$ B) $a \cdot a , = b \cdot b , = c \cdot c ,$
C) $a \cdot a , = b \cdot b , " c \cdot c ,$ D) $a \cdot a , = -b \cdot b ,$
- Q.46.** 'ù&"ò ò " Ò " ò " & 2 ò2 " 9"2Â \$"2 8'é@'Y0'2 ("é/ 'Y> 'YH"%> "TM2 "TMK'y>?
[English]: [If $a \cdot a , = b \cdot b , " c \cdot c ,$, the system of equations has:
A) •ù •Y&"Ù5"ù\$" / "TM2 B) 'YK•, 9"" ("TM@•" „æð 6öÇWF—öâ Ò 8'é>•)\$" 0"y "é ••
C) •Y(•)\$ •Y("y "TM2 D) 'iK "TM2
- Q.47.** "%o. "é 'I0 "iM" G'9@ (AP): 2, 7, 12, ... 'Y> 10"Y>• *b "Ù/"â 9"1 "ãð
[English]: [Find the 10th term of AP: 2, 7, 12, ...:
A) 45 B) 47
C) 50 D) 52
- Q.48.** "%o. "é 'I0 "iM" G'9@ 'YG '©M" %â â *iK•" "r /"1 '12 'Y> "%B'IM" "Ù/"â 9"fð
[English]: [Formula for sum of first n terms of an AP:
A) $S^{\text{TM}} = n/2 [2a + (n - 1)d]$ B) $S^{\text{TM}} = n [a + d]$
C) $S^{\text{TM}} = n/2 [a - d]$ D) $S^{\text{TM}} = 2n [a + (n-1)d]$

- Q.49.** '©M" %'â *""Ù0"é "9\$ "%o 'iM'ù>'9 'Y> 'ùK'y+"" ...7VÖ öb f—'7B æ GW al numbers) 'Y?'I("â 9"1 "ãð
[English]: [Sum of first 10 natural numbers:
A) 50 B) 55
C) 60 D) 65
- Q.50.** 'É?'%oM'iA (-3, 4) 'Y>" M'I@'ò \$" "r "ù8 '©\$" 0"Ù%"é "b ... V G ant) 'éG•" 8"Ù%"ù\$ "TMH?
[English]: [In which quadrant does point (-3, 4) lie?
A) '©M" %'â B) 'iM"Y?'I@'ò ...6V6öæB V G ant: -x, +y)
C) 'IC'I@'ò D) '©\$" 0"Ù%
- Q.51.** 'éB"" , "ù("Ù&" „÷ igin) 'YG '%o?" M'iG"i>•) 'YM'ù> "TMK'IG "TMH•#ð
[English]: [Coordinates of origin are:
A) (1, 1) B) (0, 0)
C) (0, 1) D) (-1, -1)
- Q.52.** 'É?'%oM'iA P(x, y) 'Y@ 'éB"" , "ù("Ù&" f Æ ' 8"r &"0" "â 8")\$"Ù0 'YM'ù> "TMH?
[English]: [Distance of point P(x, y) from origin:
A) x + y B) " (x² + y²)
C) x² - y² D) " (x - y)
- Q.53.** 'É?'%oM'iA'9 (2, 3) 'I0 (4, 1) 'YG 'É@'¢ " &"0" "Ù/"â 9"1 " ð
[English]: [Find distance between points (2, 3) and (4, 1):
A) 2 B) 2" 2
C) 4 D) " 8 / 2
- Q.54.** 'É?'%oM'iA'9 (x •, y •) 'I0 (x ,, y ,) 'YK 'é?">%oG "Y>"G " G'i>i' "r . 'yM'òÙ, "ù("Ù&" „Ö-B× ö-çB' "r ("ù0"Ù&"y6"é 'R 9"%o :
[English]: [Mid-point coordinates of line joining (x •, y •) and (x ,, y ,):
A) ((x •+x ,)/2, (y •+y ,)/2) B) ((x •-x ,)/2, (y •-y ,)/2)
C) (x •+x ,, y •+y ,) D) (x •x ,, y •y ,)
- Q.55.** 'ù&"ò 6-â , = 3/5 "TMK, 'IK cos ;, " F â , 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: [If sin ;, Ò 2óRÂ v† B &R 6÷2 , and tan ; f ð
A) cos ;, Ò BóRÂ F â , = 3/4 B) cos ;, Ò 2óBÂ F â , = 4/5
C) cos ;, Ò RóBÂ F â , = 3/5 D) cos ;, Ò Bó2Â F â , = 5/3
- Q.56.** sin² 63° + sin² 27° 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: [Value of sin² 63° + sin² 27° is:
A) 0 B) 1
C) 2 D) -1
- Q.57.** (sec² ;, Ò F ë" ,) 'Y> 'é>', 8'iH"R "ù8'YG 'É0"é, " 9"1\$ "â 9" f ð
[English]: [Value of (sec² ;, Ò F ë" ,) is:
A) 0 B) 1
C) -1 D) 2
- Q.58.** 9 sec² A - 9 tan² A 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: [Value of 9 sec² A - 9 tan² A is:
A) 1 B) 9
C) 8 D) 0
- Q.59.** 'ù&"ò F â , = 1 "TMK, 'IK 'YK'2 , 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: [If tan ;, Ò Â `alue of acute angle ;, —3
A) 30° B) 45°
C) 60° D) 90°
- Q.60.** •ù 'é@'%">" " "é/"â "%o " •) "é 'YG 'É0"é, " 9"%o d "%oB" M'ò "â '%oM'%"', "1# (Angle of Elevation) 'YM'ù> "TMK'y>?
[English]: [Height of tower equals its shadow length. Angle of elevation of Sun:
A) 30° B) 45°
C) 60° D) 90°
- Q.61.** "YC'IM'B " 8'É8"r , <" " 5"â „ÆöævW7B 6†÷&B öb 6—&6ÆR' "Ù/"â "TM2"é\$" 9" f ð
[English]: [The longest chord of a circle is called:
A) 'IM" ?'ÉM'ù> (Radius) B) "YM'ù>, „F— ÖWFW"•
C) '©>'¢ „ &2• D) "%oM'©0"Ù6 " G'i>

- Q.62.** 'Y?'%@ 'YC'IM'B * " , "é9"Ù/ 'É?'%M'iA (External point) "%G •Y"ù 'I. 'Y?'I(" 8"Ù*" M"b 0'y "é • " " '©@ 'É> "% 'I@ "TMH•#ð
[English]: [Maximum number of tangents drawn from an external point to a circle:
A) 1 B) 2 ('iK'%K•" " 2'ém'É>•y/'é 'É0"é, " 9"1\$" 9"%)
C) •Y(•)\$ D) 0
- Q.63.** 'ù&"ò 5"9\$"Ù\$ 'Y@ 'IM" ?'ÉM'ù> r "TMK, 'IK •Y0"Ù"YC'IM'B "â * " ?'é>'¢ ... erimeter of Semicircle) 'YM'ù> "TMK'y>?
[English]: [Perimeter of a semicircle of radius r:
A) < B) < " 2 " " Ò "fÀ + 2)
C) 2< D) < + " ò
- Q.64.** 'iK "YC'IM'IK•" "r "Ù7"y\$"Ù0'12"1 'Y> •Y(" * "é\$ 4 : 9 "TMH–B "% " \$"Ù0"ù "Ù/'é •" "â '%A'©>'B "Ù/'â 9"1 "ãð
[English]: [Ratio of areas of two circles is 4:9. Ratio of their radii is:
A) 2 : 3 B) 16 : 81
C) 4 : 9 D) 1 : 2
- Q.65.** •ù 'K", " M'y "12"r "â " 2 '©C"yM' @'ò "Ù7"y\$"Ù0'12 (Total Surface Area of Hemisphere) 'YM'ù> "TMK'I> "TMH?
[English]: [Total surface area of a solid hemisphere of radius r:
A) 2< + B) 3< +
C) 4< + D) 2/3 < +0
- Q.66.** •ù 'yK"G 'Y@ 'IM" ?'ÉM'ù> 'iK'yA'%@ 'Y0 'i@ 'É>•òÂ \$"2 "% "â 'ù\$, "ù\$"%G 'yA'%> "TMK 'É>•ù "ãð
[English]: [If radius of a sphere is doubled, its volume becomes:
A) 2 'yA'%> B) 4 'yA'%>
C) 8 'yA'%> (V " r³) D) 16 'yA'%>
- Q.67.** •ù 'ÉG") (Cylinder) 'Y> "Y "Ù0 '©C"yM' @'ò "Ù7"y\$"Ù0'12 'Y> "%B'IM" "Ù/'â 9"1fð
[English]: [Curved surface area of a cylinder of radius r and height h:
A) < +&€ B) 2< &€
C) 2< "+" ², • D) 1/3 < +&€
- Q.68.** "i 'YA (Cone) 'YG •i/'I('Y> "%B'IM" "Ù/'â 9"1fð
[English]: [Volume of a right circular cone:
A) < +&€ B) 1/3 < +&€
C) 2/3 < +0 D) < I
- Q.69.** 'YG•&"Ù0" / '©M" 5"9\$"Ù\$"ò "r \$" ('é>'© "1 'é>'yM'ò „ÖV â'Â . "é"Ù/'R „ÖVF– â' " " , "TM A") (Mode) 'éG•" 8"TM@ •i(" -"Y?'R 8•)
[English]: [Empirical relationship between Mean, Median and Mode:
A) 'É9" 2'R Ò 2 . "é"Ù/'R Ò " . "é"Ù/
B) 'É9" 2'R Ò " . "é"Ù/'R Ò 2 . "é"Ù/
C) 'é>'yM'ò Ò 2 . "é"Ù/'R Ò , "TM A") D) 'é>'yM'ù = 3 'É9" 2'R Ò " . "é"Ù/
- Q.70.** •i 'Y"1 2, 3, 5, 3, 7, 3, 8 'Y> 'É9" 2'R „ÖöFR' "Ù/'â 9"1 "ãð
[English]: [Find mode of data: 2, 3, 5, 3, 7, 3, 8:
A) 2 B) 3 ("0"Ù5"é"ù 'É>" 'ù>)
C) 5 D) 7
- Q.71.** •ù '%?'iM'©?'B 'ù("â ...7W&R W`ent) 'Y@ '©M" >'ù?'Y\$"â ... &ö& &–Æ—G" "ù\$"%@ "TMK'I@ "TMH?
[English]: [Probability of a certain/sure event is:
A) 0 B) 1
C) 0.5 D) -1
- Q.72.** 'Y?'%@ '%o'%> 'Y@ '©M" >'ù?'Y\$"â „R' "â . "é('Y?'%o"r , " "TMK'I> "TMH?
[English]: [Value of probability P(E) always lies between:
A) -1 'I0 1 B) 0 'I0 1 'YG 'É@'¢ f &B „R' &B •
C) 1 'I0 10 D) 0 'I0 •Y(•)\$
- Q.73.** 'ù&"ò „R' Ò ã R 9"2Â \$"2 tR ("TM@•"r ... †æ÷B R" " * "Ù0"é/"ù 'I> 'YM'ù> "TMK'y@?
[English]: [If P(E) = 0.05, what is P(not E)?
A) 0.05 B) 0.95
C) 0.50 D) 1.05
- Q.74.** •ù '©>%G 'YK •ù 'É>" + "y 'Y("r * " 'Ù>'ÉM'ò 8•) "Ù/'â ... ime number) •i("r " * "Ù0"é/"ù 'I> 'YM'ù> "TMK'y@?
[English]: [Probability of getting a prime number on throwing a die once:
A) 1/6 B) 1/2 (2, 3, 5 'YA"" 2ó6•
C) 2/3 D) 1/3

- Q.75.** 'l>"b " S" *IM'IK•" " ' M' @ 'éG•" 8"r 'R *IM'I> %?>Y>")> 'É>'l> "TMH–B 'R 'YM'Y> (Ace) •i("r " *"U0"é/"ù 'l> 'YM'ù> "TMK'y>?
[English]: [Probability of drawing an Ace from a pack of 52 cards:
A) 1/52 B) 1/13 (4/52)
C) 4/13 D) 1/26
- Q.76.** 'ù&"ò 'R 5"%M'IA 'YK 1720 'éG•" ,y %G '©0 20% 'Y> ">Ò 9"1\$"â 9",Â \$"2 5"%M'IA 'Y> 'YM" / 'éB")M'ò „6÷7B ice) 'YM'ù> "TMK'y>?
[English]: [Selling price of an article is 1720 with 20% profit. Find Cost Price:
A) 1600 B) 1580
C) 1640 D) 1500
- Q.77.** •ù "YM'ù"Ù\$"ò c "ù." ù •) "â 8"r "é\$"â 9", \$Y> 40 'Y?'é@/'% "ù> "%G •TM8" 0"é8"Ù\$"r 2"É 'l> "TMH–B *)0" /"é\$"U0"â " "%
[English]: [A man travels at 60 km/h and returns at 40 km/h. Average speed is:
A) 50 'Y?'é@/'% "ù> B) 48 'Y?'é@/'% "ù>
C) 45 'Y?'é@/'% "ù> D) 52 'Y?'é@/'% "ù>
- Q.78.** A 'Y?'%@ 'Y>'â "2 &"ù('éG•" " " "%@ 'Y>'â "2 R &"ù('éG•" " \$"â 9"%d 'iK'%K•" .ù2'Y0 •TM8 'Y>'â "2 "ù\$'%G 'i?'%K•" .
[English]: [A can do a work in 10 days, B in 15 days. Working together, days needed:
A) 5 'i?'€ B) 6 'i?'€
C) 8 'i?'€ D) 7.5 'i?'€
- Q.79.** 15,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R &" 8"r " 5" M"r "r 'YM" 5"9&"Ù""ò ,")/"é (CI) 'l0 "%>y>" # 'ÉM'ù>Â ...4" "â •)\$" "ù
[English]: [Difference between CI and SI on 15,000 at 10% for 2 years:
A) 125 B) 150
C) 1100 D) 175
- Q.80.** 'ù&"ò " (" "r .")2"Ù/ 'éG•" #RR " 5"9&"Ù""ò 9"2 "é , 'iK •ù '©0"ù5"é0 'YK •Y*%@ 'i*B .y 'Y?'l("r *")U0'l?"i\$ 'Y@ 'Y." " (" 9"1
[English]: [If price of sugar rises by 25%, by what percent must consumption reduce:
A) 25% B) 20%
C) 15% D) 16.66%
- Q.81.** '©M" %'â # 5"ù7'â *")U0"é "9\$ "% 'iM'ù>9 'Y> 'ùK'y+"" ...7VÒ òb f—7B # òFB æ GW al numbers) 'YM'ù> "TMK'y>?
[English]: [Sum of first 20 odd natural numbers:
A) 200 B) 400 (n²)
C) 420 D) 380
- Q.82.** •ù "%.É>"TM A 'IM" ?'ÙA'Â „W V–Æ FW al Triangle) 'Y@ 'ÙA'É> 6 "%G'é@ "TMH–B "% "â "Ù7"y\$"U0"12 'YM'ù> "TMK'y>?
[English]: [Side of equilateral triangle is 6 cm. Find its area:
A) 9" 3 "%G'é@² B) 18" 3 "%G'é@²
C) 36 "%G'é@² D) 6" 3 "%G'é@²
- Q.83.** 120 'é@'ù0 ") 'É@ 'ùM" G', SB "ù." ù •) "â " "é2 "%G •ù 'i 'ÙG 'YK 'Y?'l("r 8'é/ 'éG•" *"é0 'Y0"y " ð
[English]: [A 120m long train running at 54 km/h crosses a pole in:
A) 6 "%G'Y ' B) 8 "%G'Y '
C) 10 "%G'Y ' D) 12 "%G'Y '
- Q.84.** 'iK "% 'iM'ù>9 'Y> •Y(" *"é\$ 3 : 4 "TMH 'l0 •TM('Y> ")f"%f (LCM) 180 "TMH–B '%.y "%G 'lK'ù@ "% 'iM'ù> 'YM'ù> "TMK'y@?
[English]: [Ratio of two numbers is 3:4 and LCM is 180. The smaller number is:
A) 30 B) 45
C) 60 D) 15
- Q.85.** 'IM" ?'ÙA'Â "r \$" ("1 •Y("Ù\$•2 "1#"1 'Y> 'ùK'y+"" "ù\$"%> "TMK'l> "TMH?
[English]: [Sum of interior angles of a triangle is:
A) 90° B) 180°
C) 360° D) 270°
- Q.86.** "YC'IM'B " *" ?'y? 'l0 'YM'ù>“, "â '%A'©>'B 8'iH"R "Ù/"â "TM2"é\$"â 9"fð
[English]: [Ratio of circumference of a circle to its diameter is:
A) 1 B) '©>•, fÂ)
C) 'IM" ?'ÉM'ù> D) 2<
- Q.87.** 'iK "% 'iM'ù>9 'Y> 'éf"%f (HCF) 15 'l%"â 2–i8–b „Ä4Ò' 3 9"%d 'ù&"ò 'R 8•) "Ù/"â c 9",Â \$"2 &")8" @ "% 'iM'ù> 'YM'ù> "TMK'y@?
[English]: [HCF of two numbers is 15, LCM is 300. If one number is 60, find the other:
A) 75 B) 80
C) 90 D) 100

Q.88. 140 'YK •Y-“é "Ù/ 'yA'9('i ' K•" "r " #'%+"" "r 0")* 'éG•" "%8"r 2"ù "â "é 'y>?

[English]: [Prime factorisation of 140 is:

A) $2^2 \times 5 \times 7$

B) $2 \times 5^2 \times 7$

C) $2^3 \times 5 \times 7$

D) $2 \times 5 \times 7^2$

Q.89. '%? 'éM', . 'y "G 'YL', 8" 8•) "Ù/"â 'R '©0"ù."y/ "iM'ù> (Irrational Number) "TMH?

[English]: [Which of the following is an irrational number?

A) " 5

B) " 9

C) " 16

D) 22/7

Q.90. '©0"ù."y/ "iM'ù> 17 / 8 'Y> 'i6'62"R * "Ù0"%>" "%8"â 9"1 "ãð

[English]: [Decimal expansion of 17/8 is:

A) "i>•) (Terminating)

B) •Y6"é 'B "Y0"Ù\$" „æðâ×FW minating repeating)

C) •Y6"é 'B "%>"Y0"Ù\$" % ©0"ù."y/ / Non-terminating Non-repeating (Non-terminating)

D) "Y0"Ù\$" / Undetermined

Q.91. 'iM"Y?"%>'B , "TM'A'©& ax² + bx + c 'YG "iB'%M'ù>•) "1 'Y> 'ùK'y+"" f ± + ;"" "Ù/"â 9"1\$"â 9"fð

[English]: [Sum of zeroes of quadratic polynomial ax² + bx + c is:

A) -b / a

B) c / a

C) b / a

D) -c / a

Q.92. •ù 'iM"Y?"%>'B , "TM'A'©& 'ÉM'6>'B " "ù 'É?"% "r 6")("Ù/"é 'YK•" "â /"1 4 'l%"â " #'%+"" 9"f

[English]: [Find a quadratic polynomial whose sum and product of zeroes are 4 and 1:

A) x² - 4x + 1

B) x² + 4x + 1

C) x² - 4x - 1

D) x² + 4x - 1

Q.93. "%." " # '%? 'Y>'ð , 2 " ' 2 2 Ò " ' ; 2 " " 2 2 Ò "â 'R 'iM"Y?"I@'ò 9"" ...Væ—VR 6öÇWF—öâ' 9"1("r " 6" M'B "Ù/"â 9"f

[English]: [Condition for unique solution of a pair of linear equations:

A) a •/a , " b •/b ,

B) a •/a , = b •/b , = c •/c ,

C) a •/a , = b •/b , " c •/c ,

D) a •/a , = -b •/b ,

Q.94. 'É, a •/a , = b •/b , = c •/c , "TMK, 'IK "%." " # '%? 'Y>'ð "r "ù\$"%G "TM2 "TMK'IG "TMH•#ð

[English]: [When a •/a , = b •/b , = c •/c , the number of solutions is:

A) •Y(•) / •Y("y "TM2 (Infinitely many)

B) 'YK•, 9"" ("TM@•" ò æð 6öÇWF—öâ ... arallel Lines)

C) 'YG"Y2 •ù •Y&"Ù5"ù\$" / "TM2 / Exactly One Unique Solution

D) 'iK "TM2 / Two Distinct Solutions

Q.95. k 'YG 'Y?" , .é('YG ")?•ò 8'é@'Y0'2 ' , 2 7' Ò r \$'Y> 4x + ky = 12 'Y> 'YK•, 9"" ("TM@•" 9"1 "ãð

[English]: [For what value of k, the system 2x + 3y = 7 and 4x + ky = 12 has no solution?

A) 6

B) 3

C) 2

D) 4

Q.96. 'iM"Y?"%>'B 8'é@'Y0'2 <" 2 ' , 2 2 Ò "â 5"ù5"ù "Ù\$'Y0 (D) 'YM'ù> "TMK'I> "TMH?

[English]: [Discriminant of quadratic equation ax² + bx + c = 0 is:

A) b² - 4ac

B) b² + 4ac

C) 4ac - b²

D) " (b² - 4ac)

Q.97. "%." " # 2x² - 4x + 3 = 0 'YG 'éB")K•" " * "Ù0'YC'I? 'YM'ù> "TMK'y@?

[English]: [Nature of roots of 2x² - 4x + 3 = 0:

A) 'YK•, 5"é8"Ù\$"Y?"R .")2 '%9" (D < 0)

B) "Y>"%M'I5"ù •ù5•" 8'é>' , .")2 (D = 0) / Real and Equal Roots

C) "Y>"%M'I5"ù •ù5•" -ù("Ù('éB"" „B â ' ò &V Â æB F—7F—æ7B)æY•G7"éG'ò ò —' ational

Q.98. 'ù&"ò 8'é@'Y0'2 'x' Ò , 2 " Ò "r .")2 "%."é((Equal) "TMK•"Â \$"2 2 "â .é('YM'ù> "TMK'y>?

[English]: [If roots of 2x² - kx + 2 = 0 are equal, find k:

A) ±4

B) ±2

C) ±8

D) 0

Q.99. A.P. 3, 8, 13, 18, ... 'Y> 10"Y>• *'b "Ù/"â 9"1 "ãð

[English]: [Find 10th term of A.P. 3, 8, 13, 18, ...:

A) 48

B) 50

C) 45

D) 53

Q.100. '©M" %â " , *)0"Ù# "é 'YK•" f 2 " 2 2 âââ + 100) 'Y> 'ùK'y+"" "Ù/"â 9"1 "ãð

[English]: [Sum of first 100 positive integers:

A) 5050

B) 5000

C) 5150

D) 5500

Q.101. A.P. 7, 13, 19, ..., 205 'éG•" " 2 'Y?'I("r *'b 9"% ?

[English]: [How many terms are in A.P. 7, 13, 19, ..., 205?]

A) 34

B) 33

C) 35

D) 32

Q.102. %³ABC 'éG•" DR "R \$2 9"%d 'ù&"ò B Ò ãR 6ÒÂ D" Ò 2 6ÒÂ R Ò 6Ò 9"²Â \$"² T2 "â ."é('YM'ù> "TMK'y>?

[English]: [In %³ABC, DE "% BC. If AD=1.5, DB=3, AE=1, find EC:]

A) 2 cm

B) 1.5 cm

C) 2.5 cm

D) 3 cm

Q.103. 'iK "%." B'¢ \$"Ù0"ù-" "1 'Y@ 'ÙA'É>'9 'Y> •Y(" *"é\$ 4 : 9 "TMH-B '‰" r "Ù7"y\$"Ù0'12"1 'Y> •Y(" *"é\$ 'YM'ù> "TMK'y>?

[English]: [Sides of two similar triangles are in ratio 4:9. Ratio of their areas is:]

A) 16 : 81

B) 2 : 3

C) 4 : 9

D) 81 : 16

Q.104. •ù "%." 'YK'2 \$"Ù0"ù-" 'Y> 'Y0"Ù# 13 "%G'é@ 'I0 •ù 'ÙA'É> 5 "%G'é@ "TMH-B \$" 8" @ 'ÙA'É> 'Y@ ").Ù,"é 'YM'ù> "TMK'y>?

[English]: [Hypotenuse of right triangle is 13 cm, one side is 5 cm. Other side is:]

A) 12 cm

B) 10 cm

C) 11 cm

D) 8 cm

Q.105. 'É?'‰M'iA (-6, 8) 'Y@ 'éB"), "ù("Ù&" „÷ igin) "%G 'iB" @ 'YM'ù> "TMK'y@?

[English]: [Distance of point (-6, 8) from origin is:]

A) 10 'é>'IM" / 10 units

B) 14 'é>'IM" / 14 units

C) 8 'é>'IM" / 8 units

D) 6 'é>'IM" / 6 units

Q.106. 'É?'‰M'iA'9 (-5, 7) 'I0 (-1, 3) 'YK 'é?"> "%G "Y>")G " G'i>i ' "r .yM'òÙ, "ù("Ù&" "r ("ù0"Ù&"y6'é 'R "Ù/"â 9"1 'yG?

[English]: [Mid-point of line segment joining (-5, 7) and (-1, 3):]

A) (-3, 5)

B) (-2, 5)

C) (3, -5)

D) (-3, -5)

Q.107. $\sin 60^\circ \times \cos 30^\circ + \cos 60^\circ \times \sin 30^\circ$ 'Y> 'é>', "Ù/"â 9"1 "ãð

[English]: [Value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$ is:]

A) 1

B) 1/2

C) " 3/2

D) 0

Q.108. $9 \sec^2 A - 9 \tan^2 A$ 'Y> 'é>', "ù8'YG 'É0"é," 9" fð

[English]: [9 sec² A - 9 tan² A equals:]

A) 9

B) 1

C) 8

D) 0

Q.109. $\tan 65^\circ / \cot 25^\circ$ 'Y> 'é>', "Ù/"â 9"1 "ãð

[English]: [Value of $\tan 65^\circ / \cot 25^\circ$ is:]

A) 1

B) 0

C) " 3

D) 1/" 3

Q.110. •ù •©0"Ù""Ù5"é"" •)-"r " *" "é •TM8'Y@ •© '©>•, "r ,">'É0 "TMH-B 8")0"Ù/ 'Y> •TM("Ù('ù('YK'2 „ ævÆR öb VÆPvation) 'YM

[English]: [Shadow of vertical pole equals its height. Angle of elevation is:]

A) 45°

B) 30°

C) 60°

D) 90°

Q.111. 'ÙB'é? 'YG •ù 'É?•)&" 8"r ." ("é0 'YG '©>'b 8"r 3 ." " &")0" *" ." ("é0 'YG "i?'i0 'Y> •TM("Ù('ù('YK'2 3 9"%d 'é@'‰>" " •)

[English]: [Angle of elevation is 30° from 30m away. Height of tower is:]

A) 10" 3 m

B) 30" 3 m

C) 15 m

D) 20 m

Q.112. "YC'IM'B "r "ù8" ,"é9"Ù/ 'É?•)&" 8"r " '©@ 'y 'iK "%M'©0"Ù6 " G'i>'9 PA 'I0 PB 'Y@ ").Ù,"é 'ùK•" .y 'YM'ù> "% 'É 'r 9"1

[English]: [Relation between lengths of two tangents drawn from external point:]

A) PA = PB ("‰."é("TMK'I@ "TMH•"

B) PA > PB

C) PA < PB

D) PA = 2PB

Q.113. •Y0"Ù""YC'IM'B .y 'É("â "1# (Angle in a semi-circle) 'Y?'I("â 9"1 \$"â 9" fð

[English]: [Angle in a semicircle is:]

A) "%.'YK'2 f" ' ò &-v†B ævÆR f" •

B) 60°

C) 180°

D) 45°

- Q.114.** 'ù&"ò 8"Ù" M"b 0"y "â A 'lO PB •ù 'iB"%0"r 8"r f "r "1# '©0 'ÙA'Y@ "™K•"Â \$"2 " ðA 'Y> 'é>', "Ù/"â 9"1 "â „ò "y 'iM" 9", "ð
[English]: [If tangents PA and PB are inclined at 80°, find " POA:
A) 50° B) 60°
C) 70° D) 80°
- Q.115.** 'iM" ?'ÉM'ù> r "Y>")G "YC'iM'B "r " , \$"Ù0"ù "Ù/i ' "â "Ù7"y\$"Ù0'12 'YM'ù> "™K'y> 'É?"%o "â "1# ; , 9" fð
[English]: [Area of a sector of angle ;, ðb 6—&6/ER ðb adius r:
A) (, , ð 3c ' r Àr² B) (, , ð f ' r Àr²
C) (, , ð 3c ' r #Àr D) 2< +
- Q.116.** •ù "YC'iM'B " " ?y? 'lO 'YM"yG'iM" +"" 8•) "Ù/"é\$"Ù. 'R 0")* "%oG 'É0"é, " 9"%o —B " , 5"9\$"Ù\$ 'Y@ 'iM" ?'ÉM'ù> 'YM'ù> "™K
[English]: [If circumference and area of a circle are numerically equal, radius is:
A) 2 'é>'iM" / 2 units B) < . "é\$"Ù0'R ò À units
C) 4 'é>'iM" / 4 units D) 7 'é>'iM" / 7 units
- Q.117.** 'iM" ?'ÉM'ù> r "Y>")G •ù ' K" , " M'y "12"r "â 8'éM'©B" M'2 *97"Ù " / 'YM"yG'iM" +"" "Ù/"â 9"1\$"â 9" fð
[English]: [Total surface area of a solid hemisphere of radius r:
A) 3< + B) 2< +
C) 4< + D) 2/3 < +
- Q.118.** 'ù&"ò 'R "12"r " \$"Ù0"ù "Ù/"â &"1 " (" " & " "é , 'iK •™8'Y> •i/l('Y?'l("r " ("â 9"2 "é 'y>?
[English]: [If radius of a sphere is doubled, its volume becomes:
A) 8 'yA'%o> / 8 times B) 4 'yA'%o> / 4 times
C) 2 'yA'%o> / 2 times D) 16 'yA'%o> / 16 times
- Q.119.** "%o. "é('iM" ?'ÉM'ù> 'lO "%o. "é(•© '©>• , 5"é2"r 'R , "y2' , " 'R 6•) " "r 'ù\$"%oK•" "â '%oA'©>'B "Ù/"â 9"1 "ãð
[English]: [Ratio of volumes of a cylinder and cone of same radius and height:
A) 3 : 1 B) 1 : 3
C) 1 : 1 D) 2 : 1
- Q.120.** 'é>'yM'ðÂ . "é""Ù/'R " , "™A") 'YG 'É@'¢ 8"™@ •i(" —"Y?'R 8•) , •)' (Empirical Relationship) 'YM'ù> "™H?
[English]: [Correct empirical relation between Mean, Median and Mode:
A) 'É9" 2'R Ò 2 . "é""Ù/'R Ò " . "é""Ù/ / Mode = 3 Median - 2 Mean B) 'É9" 2'R Ò " . "é""Ù/'R Ò 2 . "é""Ù/ / Mode = 2 Median - 3 Mean
C) 3 'É9" 2'R Ò . "é""Ù/'R Ò " . "é""Ù/ D) 'é>'yM'ù = 3 'É9" 2'R Ò " . "é""Ù/
- Q.121.** "%o 'iM'ù>9 2, 4, 6, 8, 10 'Y> "%o. "é 'lO 'é>'yM'ð "Ù/"â 9"1 "ãð
[English]: [Arithmetic mean of 2, 4, 6, 8, 10:
A) 6 B) 5
C) 7 D) 8
- Q.122.** •i 'Yl"ÉK•" rÂ "Â Â BÂ 'Â RÂ , " . "é""Ù/"ù "â „ÖVF— â' "Ù/"â 9"1 "ð
[English]: [Median of data 7, 2, 11, 4, 9, 5, 8:
A) 7 B) 8
C) 6 D) 5
- Q.123.** 'É 'ù(2, 3, 5, 3, 7, 3, 8, 3, 9 'Y> 'É9" 2'R "Ù/"â 9"1 "ãð
[English]: [Mode of the distribution 2, 3, 5, 3, 7, 3, 8, 3, 9 is:
A) 3 B) 5
C) 7 D) 9
- Q.124.** 'Y?"%o@ •Y8•)-"R 'ù("â „—× ÷76— le Event) 'Y@ '©M" >'ù?'Y\$ "â "Ù/"â 9"1\$" 9" fð
[English]: [Probability of an impossible event is:
A) 0 B) 1
C) 0.5 D) -1
- Q.125.** •Y "Ù " \$" 9 '1G•) " •, S" *iM'iK•" " \$"é6 'Y@ 'y!"Ù!" . "y "%oG •ù '©\$"Ù\$ "â ("ù "é2"â "é\$"â 9"%o d •ù ")>"" 0•) 'Y> 'É>'i6"é6
[English]: [Probability of drawing a red king from a deck of 52 cards:
A) 1 / 26 B) 1 / 52
C) 1 / 13 D) 2 / 13
- Q.126.** •ù 'Y> " 8"r " '\$'R c "ù. " ù •) "â 8"r "é\$" 9", '\$'Y> 40 'Y?'é@/'%o 'ù> "%oG ")L'ù\$" 9"%o d '©B" @ 'ù>'iM" > 'Y@ 'l8'B "é2 'YM'
[English]: [Car travels at 60 km/h and returns at 40 km/h. Average speed is:
A) 48 'Y?'é@/'%o 'ù> B) 50 'Y?'é@/'%o 'ù>
C) 45 'Y?'é@/'%o 'ù> D) 52 'Y?'é@/'%o 'ù>

Q.127. 15,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R &" 8"r " 5" M"r "r 4' " 4' "â •)\$" "Ù/"â 9"1 "ãð

[English]: [Difference between CI and SI on 15000 at 10% for 2 years:

- A) 150 B) 125
C) 1100 D) 175

Q.128. A 'Y?'%@ 'Y>" M'ð "² " &"ù('éG•" \$'Y> B 18 'i?', .y '©B" > 'Y0'I> "™H–B &"1("1 'é?) " "ù\$'%G 'i?', .y '©B" > 'Y0"y 'yG?

[English]: [A does work in 12 days, B in 18 days. Together in how many days?

- A) 7.2 'i?', ò rã" F ys B) 8 'i?', ò , F ys
C) 6 'i?', ò b F ys D) 7.5 'i?', ò rãR F ys

Q.129. 'ù&"ð " (" "r .)2"Ù/ 'éG•" #RR " 5"9&"Ù"ð 9"² "é , 'IK 'yC"™?"9@ 'YK 'i*"B .y 'Y?'I("r *"Ù0'I?"i\$ 'Y@ 'Y." " (" 9"1 " \$"é "ð

[English]: [Price of sugar rises by 25%. Reduction in consumption to keep expenditure constant:

- A) 20% B) 25%
C) 16.66% D) 15%

Q.130. 'ùB'YM")?' 5"ù–"é ' , *"Ù0'éG'ù?'Y> a = bq + r 'éG•" 6"y7'12 r 'YG ")?•ð "É("%@ "i0"Ù\$ "%\$ "Ù/ "™H?

[English]: [In Euclid's division lemma a = bq + r, condition for remainder r is:

- A) 0 "d r < b B) 0 < r "d b
C) 0 "d r "d b D) r > b

Q.131. "%,%%G '1K'ù@ •Y–"é "Ù/ "% 'iM'ù> (Smallest Prime Number) 'YL', 8" 9"fð

[English]: [Smallest prime number is:

- A) 2 (•ù 'é>'IM" 8'â 'Ü>'ÉM'ð ò " „öæÇ' W'en Prime) B) 1
C) 3 D) 0

Q.132. " H'i?'R , "™A'©& (Linear Polynomial) 'Y@ •Y"ù 'I. '%>'B "ù\$'%@ "™K'I@ "™H?

[English]: [Degree of a linear polynomial is:

- A) 1 B) 2
C) 0 D) 3

Q.133. 'ù&"ð &"Ù5"ù "é\$ 'É9" *b <" Ò 7, ² " "r 6")("Ù/"é 'R ± 'I0 , " 9"1 , 'IK 1/; ² ó² 'Y> 'é>', "Ù/"â 9"1 "ãð

[English]: [If ; $\hat{A}^2$ are zeroes of $x^2 - 3x + 2$, find $1/;^2 \hat{O}^2$:

- A) 3 / 2 B) 2 / 3
C) -3 / 2 D) 1

Q.134. 'ù&"ð &"² 0"y "é •" 'R , "ù 'iA '©0 '©M" \$"ù "Ù "y& 'Y0'I@ "™H•"Â \$"² 5"" , "ù 'iA 'iK'%K•" 8'é@ 'Y0'9K•" "â "%8"â 9"" 9"1 "ãð

[English]: [If two lines intersect at a point, that point is:

- A) •Y&"Ù5"ù\$" / "™2 (Unique Solution) B) •Y(•)\$ •Y("y "™2 / Infinitely Many Solutions
C) 'YK•, 9"" ("™@•" ò æð 6öÇWF–öâ ... arallel Lines) D) 'iK "™2 / Two Distinct Solutions

Q.135. 'ù&"ð , ² ' Ò B \$'Y> x - y = 4 "™K, 'IK x 'I0 y 'YG 'é>', "Ù/"â 9"1 'yG?

[English]: [Solve x + y = 14 and x - y = 4:

- A) x = 9, y = 5 B) x = 8, y = 6
C) x = 10, y = 4 D) x = 7, y = 7

Q.136. 'iM"Y?'%>'B 8'é@ 'Y0'2 "â . "é('R 0")* (Standard Form) 'YM'ù> "™K'I> "™H?

[English]: [Standard form of quadratic equation is:

- A) $ax^2 + bx + c = 0$ ('É9"é a"" 0) B) $ax + b = 0$
C) $ax^3 + bx^2 + cx + d = 0$ D) $ax^2 + c = 0$

Q.137. "%." " # $x^2 - 9 = 0$ 'YG 'éB"" "Ù/"â 9"1 'yG?

[English]: [Roots of equation $x^2 - 9 = 0$ are:

- A) ±3 B) 9
C) 3 D) ±9

Q.138. A.P. -10, -6, -2, 2, ... 'Y> "%>" M"R •)\$" †B' "Ù/"â 9"fð

[English]: [Common difference of A.P. -10, -6, -2, 2, ... is:

- A) 4 B) -4
C) 2 D) -2

Q.139. '©M" %'â â 5"ù7'â *"Ù0"é "9\$ "% 'iM'ù>'9 'Y> 'ùK'y+"" ...7VÖ öb f—'7B â öFB æ GW al numbers) 'YM'ù> "™K'I> "™H?

[English]: [Sum of first n odd natural numbers is:

- A) n^2 B) $n(n+1)$
C) 2n D) $n(n+1)/2$

Q.140. 'É?'%M'IA (-4, -5) 'Y?' , 'IA' M'Y>•)6 'éG•" 8"Û% "ù\$ "™H?

[English]: [In which quadrant does point (-4, -5) lie?

A) 'IC'I@'ò 'IA' M'Y>•)6 (Third Quadrant)

B) '©M" %'â 'IA' M'Y>•)6 / First Quadrant (+, +)

C) 'iM"Y?'I@'ò 'IA' M'Y>•)6 / Second Quadrant (-, +)

D) '©\$" 0"Û% '©\$" 0"Û%"é "b ò "ourth Quadrant (+, -)

Q.141. 'ù&"ò \$" ('É?'%M'IA "% " G'i@ (Collinear) "™K•"Â \$"² '‰8"r , '‰("r 5"é2"r \$"Û0"ù-" 'Y> 'YM"yG'IM" +"" "Û/"â 9"¹ "ãð

[English]: [Area of triangle formed by three collinear points is:

A) 0

B) 1

C) -1

D) •Y* " ?'é?'B 0")* "‰G •Y("y / Infinitely Many

Q.142. 'IM" ?'ÛA'Â "r 6" 0"Û7 (x •, y •), (x „, y „), (x f, y f) "™K•"Â \$"² "y("Û&"Û0'R "r ("ù0"Û&"y6"é 'R "Û/"â 9"¹ 'yG?

[English]: [Coordinates of centroid of triangle with given vertices:

A) ((x •+x „+x f)/3, (y •+y „+y f)/3)

B) ((x •+x „)/2, (y •+y „)/2)

C) (x •+x „+x f, y •+y „+y f)

D) (x •x „, x f, y •y „, y f)

Q.143. 'ù&"ò 6-â „ = 3/5 "™K, 'IK cos „, "â .“é('YM'ù> "™K'y>?

[English]: [If sin „, Ò 2óRÂ f-æB 6÷2 „:

A) 4 / 5

B) 5 / 4

C) 3 / 4

D) 4 / 3

Q.144. tan 45° 'Y> 'é>', "Û/"â 9"¹\$ "â 9"¹fð

[English]: [Value of tan 45° is:

A) 1

B) 0

C) 1/" 3

D) " 3

Q.145. 1 + cot² „, "ù8'YG 'É0"é, " 9"¹\$ "â 9"¹fð

[English]: [1 + cot² „, —2 W V Â Fó

A) cosec² ;€

B) sec² ;€

C) tan² ;€

D) sin² ;€

Q.146. "YC'IM'B "r 'R „ù 'IA '©0 'Y?'I(" 8"Û* "M"b 0"y "é •" " '©@ 'É> "‰ 'I@ "™H•#ð

[English]: [How many tangents can be drawn at a point on a circle?

A) 'YG"Y2 1 (Only one)

B) 2

C) •Y(•)\$ / Infinity (")

D) 0

Q.147. •ù "YC'IM'B " "ù\$'‰@ "‰. "é 'IO "‰M'©0"Û6 " G'ï>•ù (Parallel Tangents) "™K "‰ 'I@ "™H•#ð

[English]: [Maximum number of parallel tangents a circle can have:

A) 2

B) 4

C) 1

D) •Y(•)\$ / Infinity (")

Q.148. 'IM" ?'ÉM'ù> r "Y>")G "YC'IM'B "r 'R 'IA' M'Y>•)6 (Quadrant) 'Y> 'YM"yG'IM" +"" "Û/"â 9"¹\$ "â 9"¹fð

[English]: [Area of a quadrant of a circle of radius r is:

A) 1/4 < +

B) 1/2 < +

C) < +

D) 2<

Q.149. 'IM" ?'ÉM'ù>9 R 'IO r (R > r) "Y>")G 'iK "‰ 'YG•)&"Û0" / "YC'IM'IK•" "r „, 'É("r 5")/ (Ring) 'Y> 'YM"yG'IM" +"" "Û/"â 9"¹fð

[English]: [Area of ring between concentric circles of radii R and r:

A) < ...+" Ò + "•

B) < ... " Ò ">

C) 2< ... " Ò "•

D) < + " ò +

Q.150. 'IM" ?'ÉM'ù> 7 "‰G'é@ 'I% "â •) "é 10 "‰G'é@ "Y>")G ")."Û,"YC'IM'I@'ò „y2', "â 'ù\$, "Û/"â 9"¹ "ãð

[English]: [Volume of cylinder with r = 7 cm, h = 10 cm:

A) 1540 "‰G'é@³ / 1540 cm³

B) 1450 "‰G'é@³ / 1450 cm³

C) 1500 "‰G'é@³ / 1500 cm³

D) 770 "‰G'é@³ / 770 cm³

Q.151. "i 'YA 'Y@ 'I?' M'ù •© '©>•, ...6/Æ çB †V-v‡B vÂr' "â 8")\$ "Û0 'YM'ù> "™H?

[English]: [Slant height of cone with radius r and height h:

A) " (r² + h²)

B) r² + h²

C) " (r² - h²)

D) r + h

Q.152. 'ÛA'É> a "Y>")G •ù "‰((Cube) 'YG 'éA'iM'ò 5"ù " M'2 „F-vöæ Â' " 2'ém'É>•, "Û/"â 9"¹ "ð

[English]: [Length of diagonal of a cube of edge a is:

A) a" 3

B) a" 2

C) 3a

D) a / " 3

Q.153. “Y0”Ù •Y 'I0“é2 10 - 25 'Y> “Y0”Ù ‘©?’™M’, „6Æ 72 Ö k) ‘YM’ù> “™K’y>?

[English]: [Class mark of class interval 10 - 25 is:

- A) 17.5 B) 15
C) 20 D) 35

Q.154. •i ‘Y!’ÉK•" "Â #RÂ RÂ ,Â 3RÂ # “â *” ?“%0 (Range) ‘YM’ù> “™K’y>?

[English]: [Range of data 12, 25, 15, 18, 35, 20 is:

- A) 23 B) 35
C) 12 D) 20

Q.155. ‘Y?’%@ ‘%?’iM’©?’B ‘ù(“â ...7W&R W`ent) ‘Y@ ‘©M“ >’ù?’Y\$“â “ù\$%@ “™K’I@ “™H?

[English]: [Probability of a sure event is:

- A) 1 B) 0
C) 0.5 D) •Y(“ù6”Ù “ù\$ / Indeterminate

Q.156. ‘iK “%?’YM’YK•" "2 ‘R 8“é% •™ “é2%G ‘©0 ‘Y. “%G ‘Y. •ù ‘©?’IM’B „ B ÆV 7B öæR †V B’ ‘%G ‘Y@ ‘©M“ >’ù?’Y\$“â ”Ù/“â

[English]: [Probability of getting at least one head when tossing two coins:

- A) 3 / 4 B) 1 / 4
C) 1 / 2 D) 1

Q.157. •ù “Y8”Ù\$” “2 “s# .”y ‘ÉG’©(”r “# R “â 2“é- “™K’I> “™H–B 5“%M’IA ‘Y> ‘YM“ / ‘éB”)M’ò „6÷7B ice) ‘YM’ù> “™K’y>?

[English]: [Selling at 1720 gives 20% profit. Cost price is:

- A) 1600 B) 1580
C) 1640 D) 1560

Q.158. (7 y u - 3 u x) ‘Y> •y “é ‘Y> •Y ‘R ”Ù/“â 9”1 “ãð

[English]: [Unit digit of (7 y u - 3 u x) is:

- A) 4 B) 6
C) 0 D) 7

Q.159. ‘Y?’I(”r 8‘é/ ‘éG•" "1 ‘y(5% “Y>“ M“y?’R 8“é“é0’2 ,”Ù/“é ‘Y@ ‘i0 “%G ‘iK’yA’%> “™K ‘É>•ù “ãð

[English]: [In what time will a sum double at 5% simple interest per annum?

- A) 20 “Y0”Ù7 / 20 Years B) 10 “Y0”Ù7 / 10 Years
C) 15 “Y0”Ù7 / 15 Years D) 25 “Y0”Ù7 / 25 Years

Q.160. •ù “%.’É>“™A ‘IM“ ?’ÙA’Â ” *”Ù0’IM’ùG’R -” “â b 8”y. 9”%d •y8‘Y> ‘YM“yG’IM“ +” “Ù/“â 9”1 “ãð

[English]: [Area of an equilateral triangle with side 6 cm is:

- A) 9” 3 “%G’é@2 / 9” 3 cm² B) 18” 3 “%G’é@2 / 18” 3 cm²
C) 36 “%G’é@2 / 36 cm² D) 6” 3 “%G’é@2 / 6” 3 cm²

Q.161. 3/4, 5/6, 7/8, 9/10 ‘éG•" 8‘É8”r ,’ <” -“ù(”Ù(‘YL’, 8” 9”fð

[English]: [Largest fraction among 3/4, 5/6, 7/8, 9/10:

- A) 9 / 10 B) 7 / 8
C) 5 / 6 D) 3 / 4

Q.162. ‘©M“ %’â S 8’â *”Ù0“é ”9\$ “% iM’ù>’9 ‘Y> ‘I8’B ”Ù/“â 9”1 “ãð

[English]: [Average of first 50 even natural numbers is:

- A) 51 B) 50
C) 50.5 D) 100

Q.163. ‘ù&“ò %â‡, Ò ‘ 2 %â‡, 2 ‘ Ò 3# 9”2Â \$”2 , “â .“é(‘YM’ù> “™K’y>?

[English]: [If 2^x(x-1) + 2^x(x+1) = 320, find x:

- A) 7 B) 6
C) 8 D) 5

Q.164. 20% ‘I0 10% ‘Y@ ‘iK ‘YM“ .“ù ‘1B’ò ...7V66W76—`e Discounts) ‘YG “%.’IA”)M’ò ‘Y2 ‘1B’ò ”Ù/“â 9”1 ” ð

[English]: [Single equivalent discount for successive discounts of 20% and 10%:

- A) 28% B) 30%
C) 25% D) 22%

Q.165. “i>•)\$ ‘É2 ‘éG•" (“é5 ‘Y@ ‘©>” “ù.” ù •) “â \$’Y> ‘y>“ > ‘Y@ ‘©>” “ù.” ù •) “â 9”%d ‘y>“ > ‘Y@ ‘i?’i> ‘éG•" “é2 ‘YM’ù> “™K

[English]: [Speed in still water 10 km/h, stream 2 km/h. Downstream speed is:

- A) 12 ‘Y?’é@/’% ‘ù> B) 8 ‘Y?’é@/’% ‘ù>
C) 10 ‘Y?’é@/’% ‘ù> D) 14 ‘Y?’é@/’% ‘ù>

Q.179. UPMSP 'ÉK" M' * * @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"Ü* ' , ' %0."y %00G 'YK* , ("™@•" ÷ F-ôâ B' æöæR öb B)W6PY2"Ü* ' b' 7F F-7F-72 %08'é 'iM'ü?"Y@ - Mean, Median, Mode of group
C) "Y?Y2"Ü* ' r ' ©M" \$'Ü/YM"r 'Y5"ä -"Ü0"é.R 5"ü "JM"ç ÷ F- (Q 2) Z-2Y2Y2F=Mean,Median,Mode of group IM u? Y@ - Mean, Median, Mode of group
Option B) Invalid or erroneous statement of Statistics ("%>."Ü"ü" ò Öv äâ
Median, Mode of grouped data)

Q.180. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TMG'IA, •y8 •Y"Ù" Option A) Authoritative scientific/mathematical princ
Statistics (%) 5) YU u 8" O @ "M S @ "A) @
[English]: For UPMSP Board Examination 2027, choose the correct answer for the topic.

A) "Y"?Y2"U* 'R' &&"-E—G' %*"U0"é("I'> - Classical theorem of probability ("&&C@M'E-é-C@R"U0"é("I'> - Classical theorem of probability)

Q.181. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TMG'IA, •y8 •Ym"Ù'É' %oG "‰ 'É' y?B 8"™ @ •™\$ "Ù\$ " ("y –@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2U* 'b' &V Å ÇÖ&W2%5'e8'U\$'Y'R 8•) "U/é" Ò gVæFÖVÜ Theorem of Archimedes of Östörme55real numbers) "U"eY"é"UæFÖVÜF Å Numbers "Y*-%M'15"ü"-%iMÜ*ü" Fundamentale Theorem of Arithmetic (principle of prime factors) é>9?R5%5"Ü "é"ü"V#ü"ü\$"/%?ü" Option A) C) Y?Y2U* 'b' &V Å ÇÖ&W2%5'e8'U\$'Y'R 8•) "U/é" Ò gVæFÖVÜ Theorem of Archimedes of Östörme55real numbers) "U"eY"é"UæFÖVÜF Å Numbers "Y*-%M'15"ü"-%iMÜ*ü" Fundamentale Theorem of Arithmetic (principle of prime factors) é>9?R5%5"Ü "é"ü"V#ü"ü\$"/%?ü" Option A)

Q.182. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TMG'IA, •y8 •Y""Ü'Ç/A%0%5%, FrayDl8700@)•T\$"U\$ " ("y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"U* 'r' 'CM" \$"U/YM'r 'Y5"â -"U0"é.R'5"ù")M"ç ò ÷ F-
B) "Y?"Y2"U* 'r' 'CM" \$"U/YM'r 'Y5"â -"U0"é.R'5"ù")M"ç ò ÷ F-
C) "Y?"Y2"U* 'R' 'olynomials ('É9" *b Ô vVôÖWG ical meaning of zeroes, relation between zeroes and coefficients) 'X>â"Y"é("Ü"l"y%â"Y>"X8"Miò" 'Y('Option B
D) "Y?"Y2"U* 'R' 'olynomials ('É9" *b Ô vVôÖWG ical meaning of zeroes, relationship between zeroes and coefficients) 'X>â"Y"é("Ü"l"y%â"Y>"X8"Miò" 'Y('Option B

Q.183. UPMSP EK' M * * * @ YM y> 2027 TM GTA, y8 -Y Y M' zero% Ge% esyp B8 B8 eoz euzs dnd coey cnd
[English]: For UPMSP Board Examination 2027, write the correct answer for this topic.

C) "Y?Y2"U* 'r' ©M "\$U/YM"r Y5"â -"Ü0"é.R 5"ù")M"t ÷ ÷ F-ôâ B' æðæR öb B)W6P Y2)"Ü* 'b' air of Linear Equations in Two Variables ('ik '©0"¹ "Y>")G "H'i Algebraic methods) 'Y>.Y "é"/."Y%"Y>.Y8'M"ö":Y(/ Option B) Invalid or error Pair of linear Equations in two variables (ik '©0"¹ "Y>")G "H'i2.B 8"é.G"H.Y)

Q.184. UPMSP 'ÉK“ M' *” @‘YM“y> 2027 “™G‘IA, •y8•Y”Ù“@C@G“M@E@V3B8™“@‘MS“S“”U“@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic: Mathematical principle of Pair of Linear Equations in

A) "Y?Y2U* 'R' V G atic Equations ("IM"?Y?%>B 8'é@'Y0'2 Ö Bg áüö formula, NaGrafioEquations(üproblems)B8'é@WÖ'2-09V'B äüö formula

Q.185. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TMG'IA, •y8•Y""Ü"(çp%G%)% É'y?B8™@•™\$"\$U\$" "(y-@

[English]: For UPMSB Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2U" b' ithmetic Progressions ("e("US\$ 6"U0Y)"É@B)th term sum of first n terms of arithmetic progressions ("X"/"é%\$S"6YUIM"É@(-OptimB\$ Progressions ("‰("é"UQ" 6"U0Y)"É@ nth term sum of first n terms of arithmetic progressions A) authoritative scientific/mathematical principle of C) Y-Y2U ("OM \$U/YM/Y5a-U0e.R5U")M+P=100%Y2VWPaVand distribution method @_lno_É-0a B ædær sum F+W6P Arithmetic Progressions ("X"/"é%\$S"6YUIM"É@(-OptimB\$ Progressions ("‰("é"UQ" 6"U0Y)"É@ nth term sum of first n terms

Q.186. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TMG'IA, •y8 •Y"™Ü'é/ %oG "%o 'É 'y?B 8"™@ •TM"\$"U\$ " "(y -@

[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic

[illegible]

Q.187. UPMSP 'EK' M @ 'YM' y> 2027 'TM G' IA, y8 'Y' U' e' / %G %G 'E' y? 'B 8' TM @ • 'TM' \$' U\$ " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2"U* 'r' ' %o."y " %Gg 'YK•, (" " @ " ò ÷ F-ôâ B' æðæR öb B)W6P "Y2"U* ' b' 6ô÷&F-æ FR vVöOWG y (" %? " MiG "i>•) 'EM'ü>é?I? - Dista
Y8'U'ü" %Y / Option B) Invalid or erroneous statement of Coordinate
C) "Y"?Y2"U* 'r' '©M" \$ "Ü/YM"r 'Y5"â -"Ü0"é.R'5"ü "JM"ç ò ÷ F-ôâ Z?÷&WvFræpöisFræmFröomöOWG y (" %? " MiG "i>•) 'EM'ü>é?I? - Dista
Geometry, (" %? " MiG "i>•) 'EM'ü>é?I? - Distance formula, section formula
v? "i> %? " / Option A) Authoritative scientific/mathematical principle of

Q.188. UPMSP 'ÉK“ M' * @'YM“y> 2027 “TM'G'IA, •y8 •Y“U“ Coordinate Geometry 00%? 4M'G'•Y'EM'U'•6?@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2U"* 'R' -çG&öGV7F-ôâ Fò Irigonometry ('IM' ?'YK'9."\$ü"ô "ä ?"©/ - Trigonometric ratios
 principle of Introduction to Trigonometry ('IM' ?'YK'9."\$ü"ô "ä ?"©/ - Trigonometric ratios
 C) "Y"?Y2U"* 'R' -çG&öGV7F-ôâ Fò Irigonometry ('IM' ?'YK'9."\$ü"ô "ä ?"©/ - Trigonometric ratios
 Introduction to Trigonometry ('IM' ?'YK'9."\$ü"ô "ä ?"©/ - Trigonometric ratios

Q.189. UPMSP 'ÉK“ M' * @'YM“y> 2027 “TM'G'IA, y8•Y“Ũ'G'G'G'“% 'É y?B 8“TM @ •TM\$“U\$“ ” (y-@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2U"* 'b' Trigonometric Identities & Applications ('IM' ?'YK9'.u\$' / '%00'U5%
B) "Y"?Y2U"* 'b' Trigonometric Identities & Applications ('IM' ?'YK9'.u\$' / '%00'U5%
C) "Y"?Y2U"* 'b' Trigonometric Identities & Applications ('IM' ?'YK9'.u\$' / '%00'U5%

Q.190. UPMSP 'ÉK“ M' * @‘YM“y> 2027 “TM G‘IA, y8•Y”Ü‘Û\$%G\$% ‘É y?B 8“TM@ •TM\$”Ü\$“ ” (“y –@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* 'r' 'OM" \$"Ü/YM'r 'Y5"ä-"Ü0"é.R5"ù")M"ç ò ÷ F—(B) ZY?>ZVÜEPhan-cha-FW2%5P\$US - Tangents to a circle, theorems on external tangents
C) "Y?"Y2"Ü* 'R' 6—&6ÆW2 %o5"9\$"US - Tangents to a circle, theorems on external tangents
D) "Y?"Y2"Ü* 'R' 6—&6ÆW2 %o5"9\$"US - Tangents to a circle, theorems on external tangents

Q.191. UPSPSP EK' M' *TM @ YM Y> 2027 *TM G I A, •y8 •Y"U"e/ %G %o E'Y" B 8 *TM @ •TM\$ "U\$ " ("y -@
 [English]: For UPSPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"U* ' ' ' %o."y %oG 'YK•, ("™@•" ÷ ÷ F—äB 'P æðæR öb B)W6PY2"U* 'b' &V2 &V/E FVB Fð 6—&6ÆW2 %o5'9S"U\$"1 %oG %o 'É 'y?Y2
C) "Y?Y2"U* 'r' '©M" \$"Ü/YM"r 'Y5"ä-"Ü0"é:R 5"ù")M"¢ ÷ ÷ F—äB 'P æðæR öb B)W6PY2"U* 'b' &V2 &V/E FVB Fð 6—&6ÆW2 %o5'9S"U\$"1 %oG %o 'É 'y?Y2
Related to Circles "YC'IM'IK•" 8"r 8•).•"ü\$ 'YM'Yç'IM" +"" 06V7F÷÷
of Areas Related to Circles ("YC'IM'IK•" 8"r 8•).•"ü\$ 'YM'Yç'IM" +"" 06V7F÷÷

Q.192. UPMSP 'ÉK" M' * * @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

Q.193. UPMSP 'ÉK" M' ** @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'J"/'É"‰G"‰"É'y?'B 8"TM @•TM\$"Ù\$" " ("y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

Q.194. UPMSP 'ÉK" M' ** @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'J'É' %oG "‰ É 'y'?'B 8"TM @ •TM\$"Ù\$ " ("y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* 'r' '©M' \$"Ü/YM'r 'Y5"ä -"Ü0"é."R 5"ù)"M"ç ò ÷ F-6@ ZY?Y2Ü*P ractica de la Estadística y Probabilidad"Ü0"é"ù 'l> - Classical theoretical probability

B) "Y?"Y2"Ü* 'R' '©&& &-Æ—G' %*"Ü0"é"ù 'l> - Invalid or erroneous statement of Probability ("©M">"ü"Y5"ä Ü 0.6Æ 76-6 Å

C) "Y?"Y2"Ü* 'R' '©&& &-Æ—G' %*"Ü0"é"ù 'l> - Classical theoretical probability (theoretical probability)

Q.195. UPMPSP 'EK' M' *'' @YM"y> 2027 "TM'G'IA, •y8•Y""Ù'"/é' %oG "‰ 'É 'y?B 8"TM @ •TM\$""Ù\$" " ("y -@

[English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Û* ' ' ' %o."y "‰G 'YK•, ("M@#" ò ÷ F-äâ B' æðæR öb B) W6"Y2"Û* 'b' &V Â çVÖ&W/2 %o5"e8"Û\$Y?"R 8•) "Û"/é• " Ò gVæF ÖVçF Â
C) "Y?"Y2"Û* 'r' '©M" \$"Û"/YM"r 'Y5"â-"Û0"é."R 5"ù")M"¢ ò ÷ F-äâ B' æðæR öb B) W6"Y2"Û* 'b' &V Â çVÖ&W/2 %o5"e8"Û\$Y?"R 8•) "Û"/é• " Ò gVæF ÖVçF Â

Q.196. UPMSP 'ÉK" M' * @'YM"y> 2027 "™G'IA, •y8 •Y"Ú' / Mathematical Principle of Real Numbers ("Y" %MMS.
[English]: For UPMSP Board Examination 2027, choose the correct answer for the following question.

A) "Y?Y2U" * 'R' polynomials ('É9" *b Ö vVÖÖWG ical meaning of zeroes relationship between ('É9" and coefficients meaning of zeroes relationship scientific/mathematical principle of Polynomials ('É9" *b Ö vVÖÖWGives meaning of coefficients) ShirYHéVbén 24 Yes and More Information B) "Y?Y2U" * 'M \$U/YM1 Y5a -00e.R5u "JM" 0 ÷ F=0 21-1-k2v6F-Van der Waals and Dalton ("É9" 9 ÷ F=0a 6 2e0eR 00 F VöP erroneous statement of Polynomials ('É9" *b Ö vVÖÖWG ical meaning of

Q.197. UPMSP 'ÉK" M' * @'YM"y> 2027 "TM'G'IA, •y8•Y"™Ù'Ér/r%GÉ%ti%ÉštyP B88v8mz8r8S8s8d c8ffici8@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2U"* 'b' air of Linear Equations in Two Variables ('iK '©0"B")Y?Y?GZ"U"R R alé @ M02ar EquatOnspInital & Algebraic ("kE00ds")Y?S"É"É"
Pair of Linear Equations in Two Variables ('iK '©0"1"Y">"G "H"2"R B6lGa V08"b"b"b"s) Graphical Method Applied to Pair of Linear Equations ('v#h"\$\$/ %>2"1" / Option A
C) Y?"Y2U" F"©M \$ U/YM r Y5 a -U0 e.R5 u)JM t o = F-02 Z 1 = k2 v2 P v ar d i s t a n c e b e t w e e n @_0 = F-02 B_2 e o d e r o f F1W6P mathematical principle of Pair of Linear Equations in Two Variables

Q.198. UPMSP 'ÉK" M' * @'YM"y> 2027 "TM'G'IA, •y8 •Y"Ù'Ú'ÉK" %G " %Éy?B'8T@ @'YMS'Ú\$Ú'Ú. (Cyp@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* 'r' '©M" \$"Ü/YM"r 'Y5"a-"Ü0"é.R 5"ù")M"t ò ÷ F- (B) 2Y?2Y2Ü*YpM" Distrac Equations ("iM"Y?"%>B 8'é@'Y0'2 0 V G atic formula

Q.199. UPMPSP EK' M' *' @YM'Y> 2027 GTA, •y8 •Y"™/100%GS% E'y?B 8"™ @ •™\$"Us" ("y-@
[English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* ' ' %..y "°G G 'YK•, ("™@•" ÷ ÷ F—ä B' æöæR öb B) W6PY2"Ü* 'b' ithmetic Progressions ("°..é"Ü\$ 6"Ü0"y"™É@ - nth term, sum of first n terms)
 C) "Y?"Y2"Ü* 'r '©M" \$"Ü"YM"r 'Y5"â -"Ü0"é.R 5"ù")M"¢ ÷ ÷ F—ä B' æöæR öb Option B) Invalid or erroneous statement of Arithmetic Progressions ("°..é"Ü\$ 6"Ü0"y"™É@ - nth term, sum of first n terms)
 D) "Y?"Y2"Ü* 'r '©M" \$"Ü"YM"r 'Y5"â -"Ü0"é.R 5"ù")M"¢ ÷ ÷ F—ä B' æöæR öb Option C) Invalid or erroneous statement of Arithmetic Progressions ("°..é"Ü\$ 6"Ü0"y"™É@ - nth term, sum of first n terms)

Q-200. UPMSP 'ÉK" M' ** @'YM"> 2027 "™G'IA, •y8 •Y™Ü" Option A: Authoritative scientific/mathematical principle
Arithmetic Progressions (%, e(US "U0y) £@@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

Q.201. UPMSP 'ÉK" M' *' @'YM"y> 2027 "TMG'IA, y8•Y"Ù"é/ %oG "%o 'É y'y'B 8"TM@ •TM\$"Û\$" " ("y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?·Y2'U"· 'b' 60÷&F-æ FR vVöÖWG y ('%0?" M'iG'i>)· É'M'ù->Bé?Y?-Y2'starReform&PlaceF&ionvöüMGay (%?Y'M'GU)>·YÉ'M'ù>Y8?M?è Dysta
Geometry ('%0?" M'iG'i>)"· É'M'ù>é2!/? · Distance formula, section fofôUü%2'ù / Option A) Authoritative scientific/mathematical principle of
C) Y?·Y2'U"· @M \$U/YM+ Y5'a-U0e.R5'u)JM'to÷ F-öü ZY2-kZvE'Fv&dist&andBottom @·@·0'÷é?l? B-aö&R öf FivöP
Coordinate Geometry ('%?· M'iG'i>)"· É'M'ù>é?l? - Distance formula, section

Q.202. UPMSP 'ÉK" M' * @'YM"y> 2027 "TMG'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?B 8"TM@ •TM\$"Û\$" " ("y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* 'r' '©M' \$"Ù/YM'r 'Y5'a -"U0é.R 5"ù)"M't ò ÷ F-ôâ ZY?ZVÜEPhanGöGV7F-ôâ Trigonometry ('IM' ?'YK'9."ü\$ò "ä *** '?'©
C) "Y?"Y2"Ü* 'R' -çG&öGV7F-ôâ Fò Trigonometry ('IM' ?'YK'9."ü\$ò Yâ Y2ÜE/-Trigonometric ratios & standard values below B) Invalid or error
principle of Introduction to Trigonometry ('IM' ?'YK'9."ü\$ò "ä *** '?'©Introduction to Trigonometry ('IM' ?'YK'9."ü\$ò "ä *** '?'©/- Trigonometric ratios

Q-203. UPMPSP 'EK' M' *' @YM'> 2027 "TM G'IA, •y8 •Y'U'Veit%G% 'E' y'7 B'8™ @ •TM\$'US' "('y -@
 [English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* ' ' ' %..y "°G G 'YK•, ("IM@•" ÷ ÷ F—ä B' æðæR öb B) W6PY2"Ü* ' b' Trigonometric Identities & Applications ("IM" ?YK'9."u\$ " / %°0"
Distances) Y> > X "é"U" / %°0"Y> > Y8'IM"ä "Y (/ Option B) Invalid or erroneous
C) "Y?"Y2"Ü* 'r '°M" \$"Ü/YM'r 'Y5'ä—"Ü0'é.R 5"ü ")M'¢ ÷ ÷ F—ä 21°—æðæR öb B) W6PY2"Ü* ' b' Trigonometric Identities & Applications ("IM" ?YK'9."u\$ " / %°0"
Trigonometric Identities & Applications ("IM" ?YK'9."u\$ " / %°0" C) Invalid or erroneous

[illegible]

A) "Y?Y2U* 'R' 6—&6/EW2 %5"9\$ "Ü\$ - Tangents to a circle, theorems on external point tangents to a circle, theorems on internal point tangents to a circle, theorems on the principle of Circles ("YC:IMB Ö Tangents to a circle, theorems on external point tangents to a circle, theorems on internal point tangents to a circle, theorems on the principle of Circles")

Q.205. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y? Y2U" * 'b' & V 2 & V/E FVB Fo 6—& 6/EW2 %o%9s"ù\$¹ %m-%k² E2y0BRÚV\$S ÚV/V FVEFor Sand&Eym2his ú\$úšle%G %%"É"V?P+2
to Circles ("YC IM JK."8"r.8.)")ù\$-YM"G IM "+"Ò 6V7F÷2 æB6w0QvCGM>b>g>92RER% "U:"á("ü"/y""ù\$/:/%?"ü /_Option A) Authoritativ
C) Y? Y2U r "CM \$ U/YMr Y5a -Üö.e.R 5 u" JM c ò = F=ing 2Z&V/E vanc dist from dition%) : ö = F-päB yem gçIM + P+H 6V7F÷2

Q.206. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM @ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2"U* 'R' 7W& ace Areas and Volumes ("C"yM' @d "U7y\$U0'12 'I0 .

Q.207. UPMPSP EK' M' *** @YM'y> 2027™TG1A, y8-Y™U'e/%G% E'y?B8™@.•™\$U\$ " ("y-@
 [English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2"Ü* 'r' 'ÖM" \$"Ü/YM"r 'Y5"ä-"Ü0"é.R'5"ü"JM"ç ÷ F—(0 2/2—2Y2F—Mean, Median, Mode of group

Q.208. UPMSP 'ÉK" M' ** @'YM"y> 2027 "™G'IA, •y8 •Y"Ù" Option A1, Authoritative Scientific/mathematical princ-
Statistics (%) É YU B 8 0 QV AA QVP- aa Q6FFr
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"U* 'R' &ô& &—E—G' %o*"U0"é"ù 'l'> - Classical theoretical probability ('©M">"ù"Y\$â"à"ô"6"É"76—6"Â Classical/theoretical probability)

Q.209. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É'y?B 8"TM@ •TM\$"Ù\$" " ("y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"U* 'b' & V Å çVÖ&W2 %0o5'e8"\$"Y?"R 8*) "Ü"é • " ò gVöVÖVÜÅ Treedreä qv ÖxW12%5'tealüßä pñæfö) "M"ëY""è(U"æP"ÖVgF"V"ë Numbers ("Y">"%m"l5"ù,"%m"jM"ù"•H" Fundamentale Theorem of Arithmetic (primality decomposition) >é>?>P-5%"00") Ü"é("U"lY""üS"/,%ç"ü"l-Option A) C) Y?"Y2"U*"CM \$U"YM"lY5"a UO"e.P 5"u"JM"t ö = F=0a 2"2=&Vd"Evandistatund notation @:"0 - f - ba P. æqæP"öb"Flwop mathematical principle of Real Numbers ("Y">"%m"l5"ù,"%m"jM"ù"•H" Fundam

Q.210. UPMSP 'ÉK" M' ** @'YM"> 2027 "™G'IA, •y8•Y"Ù'Ù/A/%G1% Éy0B18™G)•™\$"Ú\$ " ("y-@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"Ü* 'r' '©M"\$ \$"Ü/YM'r 'Y5"a -"Ü0"é."R 5"ù")M"¢ ò ÷ F—(6) ZY2"Y2VÜEPvaoiydismatim(Espitib Ò vVoÖWG ical meaning of zeroes, relation between zeroes and coefficients), > >,"Y"é","Ü,"y%>"Y>,"Y8IM,ò,"Y"/Option B

C) "Y?Y2"Ü* 'R' polynomials ('É9)"*b Ò vVoÖWG ical meaning of zeroes),relationsho between zeroesand coefficients statement of Polynomials./Egü"ñ p Q VVoÖWG ical meaning of

Q.211. UPMSP EK' M *¹ @YM y> 2027™ GtA, y8-Y™Y™ zero%GetaDesypB8Beo@ezmsd coefy.ced
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

C) "Y"?Y2"U* 'r' '©M' \$"Ü/YM'r 'Y5'a- "Ü0'é.R 5"ù)"M't ÷ ÷ F-ôâ B' æðæR öb B) W6P?Y2"Ü* 'b' air of Linear Equations in Two Variables ('ik '©0"¹ "Y>")G "H'i Algebraic methods) 'Y>. Y "é"ü"/.Y%Y">.Y8'M'ò 'Y(/ Option B) Invalid or error message displayed. The input string is not a valid mathematical expression.

Q-212. UPSPSP 'ÉK" M' *" @'YM">y> 2027 "™G'IA, •y8•Y"U"Algebraic methods y>>@M">é>9R'S"º"U"eI
[English]: For UPSPSP Board Examination 2027, choose the correct answer for this topic: é@YQ?&" "l) - Graphical

A) "Y?Y2Ü* 'R' V G atic Equations ('iM:"Y?"%> B'8'e@'Y0'2 Ò BG ærð2tñula, NaGæriofEquatíons(íipòBé@W'0-2ó97'B áib6 fúrtne principle of Quadratic Equations ('iM:"Y?"%> B'8'e@'Y0'2 Ò V G æirðmòrle pñstüæe o' tñætt's wöyñ çrablé gñis) 0- Y'/(Option B) Invalid or erroneous C) Y?Y2Ü* 'R' V G atic Equations ('iM:"Y?"%> B'8'e@'Y0'2 Ò V G æirðmòrle pñstüæe o' tñætt's wöyñ çrablé gñis) 0- Y'/(Option B) Invalid or erroneous

Q.213. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM G'IA, •y8 •Y"Ù'16p%G% "É y?B 8"TM @ •TM\$"Ù\$ " "(y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2U* 'b' ithmetic Progressions ("%."é"Ú\$ "6"Ú0"Y"É@ B) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ C) "Y?Y2U* 'b' ithmetic Progressions ("%."é"Ú\$ "6"Ú0"Y"É@ D) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ E) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ F) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ G) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ H) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ I) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ J) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ K) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ L) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ M) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ N) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ O) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ P) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ Q) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ R) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ S) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ T) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ U) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ V) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ W) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ X) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ Y) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@ Z) nth term sum of first n terms of arithmetic progressions (("%."é"Ú\$ "6"Ú0"Y"É@

Q.214. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ "°G "°E "y?B 8"TM @ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2"Ū* 'R' Triangles ('IM' ?'ŪA'Ō 6-Ō-Æ ity criteria, Basic Proportionality Theorem/Thales')
 D) "Y?Y2"Ū* 'R' Triangles ('IM' ?'ŪA'Ō 6-Ō-Æ ity criteria, Basic Proportionality Theorem/Thales')

Q-215. UPSP/EK/M^{*}/M^{**}@YM/y>2027™G'IA,y8•Y™U/e%‰G‰‰e'y?B8™@.™\$™Us "™y-@
[English]: For UPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"Ü* ' , ' %0. 'y "0%G 'YK•, ("™@•" ò ÷ F-ôâ B' æðæR öb B' V6P'Y2"Ü* ' b' 6ö÷&F-æ FR vVÖWVG y ("0%?" M'iG"i>•) 'ÉM'ù>é'?! - Dista
 C) "Y?Y2"Ü* 'r '©M' \$'Ü/YM'r 'Y5'â- "Ü0'é. R'5"ù "JM'ç ò ÷ F-ôâ Z'÷-ZV&F-æFRvVÖWVG y ("0%?" M'iG"i>•) 'ÉM'ù>é'?! - Dista

Q.216. UPSP 'ÉK" M' *" @'YM">y> 2027 "™G'IA, •y8•Y""Ù"/
[English]: For UPSP Board Examination 2027, choose the correct answer for this topic.
Coordinate Geometry (%)> M1G1>Y"EM'u's>?>

A) "Y?Y'2U* R' -çG&ôGV7F-ôâ Fô Trigonometry ("IM"?YK'9."ùB)ôYAYZTÜ/bTrigG&ôGV7F-ôâ 50cTrigonometry(üB)"YKC9M'u\$ê>"R'5C% principle of introduction to Trigonometry ("IM"?YK'9."ù\$ò"à""?"S&aTaTrigonometryYatôs"/&S&i/aTrigonometryY6S)M'è"X(/ Option B). Invalid or error

C) YP'YZU' r' OM \$ U'YM! Y5a -UO.e.R5 u' JM c ô- F-ôâ ZT=V&F=F&aInstr&to BeMon (/ option C)-F-pa B. e o&E ob FTW6P
Introduction to Trigonometry ("IM"?YK'9."ù\$ò"à""?" /option D)- Trigonometric ratios

Q.217. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM G'IA, •y8 •Y"Ù'Ver%G "% É'y?B 8"TM @ •TM\$"Ù\$ " ("y –@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2Ü* 'b' Trigonometric Identities & Applications ('IM' ?YK'9 "ü\$"/%0") 15 Distances, Ys, CMC, Y?g, Q'D,5%, YV, YG/D, YF, Y7F, and Q'2) Option A) Author
C) Y?Y2Ü* 'b' Trigonometric Identities & Applications ('IM' ?YK'9 "ü\$"/%0") 15 Distances, Ys, CMC, Y?g, Q'D,5%, YV, YG/D, YF, Y7F, and Q'2) Option A) Author

principle of trigonometric identities & Applications (IM 7.1K 9.0\$ / %0.0\$ % Distances)

Q-218. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü*' r' ©M "\$"Ü/YM'r 'Y5'a -"U0ë.R'5"ù)"M'c ò ÷ F—(B) 2Y?Y2ÜEPhan—&6/EW2%59\$"—Tangents to a circle, theorems on external tangents)
C) "Y?"Y2"Ü*' R' 6—&6/EW2 %5\$"—Tangents to a circle, theorems on external tangents)
D) "Y?"Y2"Ü*' R' 6—&6/EW2 %5\$"—Tangents to a circle, theorems on external tangents)

Q.219. UPSPSP EK* M @YM y> 2027 TM GIA, y8 YU e/ %G %E y? B8 TM @ •TM\$ "Ü\$ " ("y -@
 [English]: For UPSPSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2"Ü* 'r' '©M" \$"Ü"YM"r 'Y5"â "Ü0"é. 'R 5"ù "M"¢ ò ÷ F—ôä B' æðæR öb B)W6PY2"Ü* 'b' &V 2 &V/Æ FVB Fò 6—&6ÆW2 %005'9\$"Ü\$1 %00G %00 'É 'Y?'F

Q-220. UPSP 'ÉK“ M' ** @‘YM“y> 2027 “TM’G‘IA, •y8•Y”Ü“
[English]: For UPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y?2"U* 'R' 7W&ace Areas and Volumes ('C"C"yM" @ @ "U7B\$%0Y2"U*'#I(7Solids combination volumes('C"C"yM" @ @ "#J?97P052%0U
of Surface Areas and Volumes ('C"C"yM" @ @ "U7B\$%0Y2"U*'#I(7Solids combination volumes('C"C"yM" @ @ "#J?97P052%0U
C) Y?Y2U* 'R' 7W&ace Areas and Volumes ('C"C"yM" @ @ "U7B\$%0Y2"U*'#I(7Solids combination volumes('C"C"yM" @ @ "#J?97P052%0U

Q.221. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$ " ("y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2"U* 'b' 7F F—7F—72 %8"e "IM"ù"?Y@ - Mean, Median, Mode of grouped data—7F—72%8"e "YM"ù"?Y@ - Mean, Median, Mode of grouped data

Q.222. UPMSP 'ÉK" M' ** @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ("y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2Ü* 'R' && &-E-G %*"Ü0'é"/ù 'I> - Classical theoretical probability (invalid or erroneous statement of Probability) (C)M>ü?Y?ä Ü 6/E 76-6 Ä

Q.223. UPMPSP EK' M' * * * @ YM"y> 2027 "TM'G'IA, •y8 •Y"Ù'/'é/ %oG %' 'É 'y?B 8"TM @ •TM\$"Ù\$ " ("y -@
[English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2"Û* ' ' %..y " %G 'YK•, ("™@* " ÷ ÷ F-ðâ B' æðæR öb B)W6PY2"Û* 'b' & V Â çVÖ&W'2 %5"e8"Û\$Y"?R 8•) "Û"é • " ò gVæF ÖVçF Â
irrational proofs) X> Y "é"Û, YX"Y> Y8"U"9 "Yç (Option B) Invalid or error
C) "Y"?Y2"Û* 'r' '©M" \$ "Û"YM"r 'Y5"â -"Û0"é.R 5"ù ")M"ç ÷ ÷ F-ðâ B' æðæR öb B)W6PY2"Û* 'b' & V Â çVÖ&W'2 %5"e8"Û\$Y"?R 8•) "Û"é • " ò gVæF ÖVçF Â
Real Numbers) X> Y "é"Û, YX"Y> Y8"U"9 "Yç (Fundamental Theorem of Arithmetic)

Q.224. UPMPSP 'ÉK" M' * @'YM"> 2027 "TM'G'IA, •y8 •Y"Ú' / (Mathematical principle of Real Numbers ("Y" > %0M))
 [English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic(s)

A) "Y?Y2Ü" *R' olynomials ('É9") *b Ô vVôÖWG ical meaning of zeroes relations between (Zeros and coefficients) meaning of zeroes relation scientific/mathematical principle of Polynomials ('É9") *b Ô vVôÖWG ical meanings of terms/coefficient(s) b/sharing between Zeros and Coefficients B) Y?Y2U" *C M \$ U/YM† Y5 a - U0 e R5 u)M† 0 ÷ F-10 a 27÷ k2v6F Vane distribution function ("É9") *b Ô vVôÖWG ical meaning of erroneous statement of Polynomials ('É9") *b Ô vVôÖWG ical meaning of

Q.225. UPSP 'ÉK' M' *^{*}@'YM"> 2027 "™G'IA, •y8•Y™Ù'Û'ËrÚsGéñíofstýp'BègWècèzúsgúsd cœfficiêd
[English]: For UPSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2U"? 'b' air of Linear Equations in Two Variables ('ik '©@")YY7G2"H"TR alé @ M0eR EquatOnspitTwoVargelssed ('ke00ds)"Y">E""6("Pair of Linear Equations in Two Variables ('ik '©@" V>")G " H'i2' R Belpay082b'ubs) G'raohMaCh'AggeB5'"m&i'teds)'"/%:"\$"/ %>2'u / Option A

C) Y?Y2U" F-OM \$ U/YM I-Y5d-U0'e.R5'u JM C0 ÷ F= -0.21÷kzvev'Evaadistrahng'gofion' @_0 =F-uB'eeoER 0 F1W6P mathematical principle of Pair of Linear Equations in Two Variables

Q.226. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM'G'IA, •y8 •Y"Ù'ÉK"%G %"Éy?B8T@'@'YQ\$U\$Ù." (Cyp@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

Q.227. UPMPSP EK' M' @ YM'y> 2027' TM GTA, y8-Y''M'10p%G% E'y?B 8''M @ •TM\$'US' "(y-@
[English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?"Y2"Ü* 'r' '©M" \$"Ü"/YM"r 'Y5"â - "Ü0"é. 'R 5"ù ")M"¢ ÷ ÷ F—öä B' æðæR öb B)W6"Y2"Ü* 'b' ithmetic Progressions ("‰. "é"Ü\$ "6"Ü0"y"™É@ - nth term, sum of first n terms)

Q.228. UPMSP 'ÉK" M' *" @'YM"y> 2027 "™G'IA, •y8 •Y"Ù" Option A) Authoritative scientific/mathematical principle
Arithmetic Progressions (%. e" US "U0"y"É@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2U* 'R' Triangles ('IM' ?UAÂ Ô 6-Ö-E ity criteria, Basic Proportionality Theorem/TMAes) YA-ÖÖM Ö-E'9y R5ka, Bas (U P r p u s h o h principle of Triangles ('IM' ?UAÂ Ô 6-Ö-E ity criteria, Basic Proportionality Theorem/TMAes) Y8:IM'â 'Y./ Option B) Invalid or erroneous statement
C) Y?Y2U* 'R' Triangles ('IM' ?UAÂ Ô 6-Ö-E ity criteria, Basic Proportionality Theorem/TMAes) YA-ÖÖM Ö-E'9y R5ka, Bas (U P r p u s h o h principle of Triangles ('IM' ?UAÂ Ô 6-Ö-E ity criteria, Basic Proportionality Theorem/TMAes) Y8:IM'â 'Y./ Option B) Invalid or erroneous statement

Q.229. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ "°oG "°o 'É'y?'B 8"TM @ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"U" 'b' 60÷&F-æ FR vVöOWG y ("%?" M'iG"i>•) 'ÉM'ù>B9?Y?"Y2"U" 'b' 60÷&F-æ FR vVöOWG y ("%?"Y M'iG"i>•)Y'ÉM'ù>Y8?M'É D'YtA
Geometry ("%?" M'iG"i>•) 'ÉM'ù>é2!?' Distance formula, section for U (%)2'ù / Option A) Authoritative scientific/mathematical principle of
C) Y?"Y2"U" 'b' 60÷&F-æ FR vVöOWG y ("%?" M'iG"i>•) 'ÉM'ù>B9?Y?"Y2"U" 'b' 60÷&F-æ FR vVöOWG y ("%?"Y M'iG"i>•)Y'ÉM'ù>Y8?M'É D'YtA
Coordinate Geometry ("%?" M'iG"i>•) 'ÉM'ù>é2!?' Distance formula, section

Q.230. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ "°°G "°° 'É'y?B 8"TM @ •TM\$"Ù\$" " ('y-@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2"Ü*" 'r' '©M' \$"Ü"/YM"r 'Y5"a -"Ü0"é."R 5"ù")M"¢ ò ÷ F-âä Fò Trigonometry ('IM' ?'YK'9."ù\$"ò "ä *?"'©/ standard values) 'Y> .Y"é"/U."Y%"> .Y8"IM"ò ."Y"/ / Option B) Invalid or error

C) "Y"?Y2"Ü*" 'R' -çG&öGV7F-âä Fò Trigonometry ('IM' ?'YK'9."ù\$"ò "ä *?"'©/ - Trigonometric ratios & standard values) 'Y> .Y"é"/U."Y%"> .Y8"IM"ò ."Y"/ / Option B) Invalid or error

principle of Introduction to Trigonometry ('IM' ?'YK'9."ù\$"ò "ä *?"'©/ - Trigonometric ratios & standard values)

Q-231. UPMSP 'ÉK" M' * * @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2Ü" ' ' %oo"y "ooG 'YK•, ("™@•" ò ÷ F—ôâ B' æôæR öb B) W6PY2"Ü" ' 'b' Trigonometric Identities & Applications ("IM" ?YK'9."u\$ / "%oo0 Distances) Y>Y"é("Ü) Y%("Y>Y8IM"ò Y"/ (Option B) Invalid or erroneous C) "Y?Y2Ü" 'r' '©M" \$"Ü/YM"r 'Y5"â—"Ü0"é.R'5"ù "M"ç ò ÷ F—ôâ Z?—&ZUEP"Trigonometric Identities & Applications ("IM" ?YK'9."u\$ / "%oo0 U5g."u\$ / "%oo0 Trigonometric Identities & Applications ("IM" ?YK'9."u\$ / "%oo0 U5g."u\$ / "%oo0

232. UPMPSP 'ÉK“ M' *“(‘YM“y> 2027 “TMG‘IA, •y8•Y“Ú‘E yB8“TM“U“•T.M.S.“U“S“Y“@
[English]: For UPMPSP Board Examination 2027, choose the correct answer for this topic.

Q-232. UPMSPP 'ÉK" M' *⁰ @'YM">y> 2027 "™G'IA, •y8•Y"Ü"U" / principle of Trigonometric Identities & Applications
[English]: For UPMSPP Board Examination 2027, choose the correct answer for this topic.

[illegible]

233. UPMSP 'ÉK" M' *" @'YM"> 2027 "TMG'IA, •y8 •Y"Ù'ér/ %oG %"o 'É y?B 8"TM @ •TM\$"U\$" " ("y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"Ü* 'b' &V 2 &VÆ FVB F0 6—&6/EW2 %5"9\$"Ü\$*1 "%m%2"E2Y0BRÜZV\$2 007E F5CFor Sand Segments 15081 dble"%G %%"E"Y2P
to Circles ("C"IM"K": g'r 8:•)"ü\$ YM"YG"IM:"""0 JM6VF7F:2 æB6H0Y0C2001>>6>928ERK:"U"æ"ü"/yH"us"/:%?ü./ Option A) Authoritative
C) Y?Y2Ü"r CM \$U/YMR Y5a-00e:R5U JM6F0 ÷ F=0P2Y2VÆVandistandof Region @:0÷F=0a.Bæ0erMP+P+W6P
of Areas Related to Curcles/8CM"K":g'8:•)"ü\$ YM"YG"IM:"""0 JM6VF7F:2

234. UPMSP 'ÉK" M" *" @'YM"> 2027 "TMG'IA, •y8 •Y"Ù'él' %oG %o" É'y?B 8"TM @ •TM\$U\$ " "(y-@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2"U* 'R' 7W& ace Areas and Volumes ('@C"yM' @@ "U7y\$y\$U0'2 10. conversions) Y>Y'6"U'1%Y>Y8'4M@ 'Y' / Option B) Invalid or erroneous

Q.235. UPMSB 'EK' M' xxi @YM'y> 2027 "TM G1A, y8-Y'mU'c' %ooG %oo E'y?B'8"TM @ •TM\$U\$ " ('y-@
 [English]: For UPMSB Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2Ü* 'r' '©M" \$"Ü/YM"r 'Y5"ä-"Ü0"é.R5"ü"JM"¢ ð ÷ F-0ä B' æðæR öb B)W6P2)Ü* 'b' 7F7F-7F-72 %008"é 'iM"ù?"Y@ - Mean, Median, Mode of group

236. UPMSP ÉK M * @ ^YM> 2027 TM G'IA, y8 •Y™Ü•
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

237. UPMSP 'ÉK" M" *" @'YM"y> 2027 "TMG'IA, •y8 •Y"Ù'él/ "G" "É"y?B 8"TM@ •TM\$"Ù\$" " ("y-@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"U* 'b' &V Å çVÖ&W2 %5"é8"U\$Y?"R 8•) "U"/é"é" Ò gVæFÖVçFÅ Theorem of Archimedes (real number principle) "U"éY"é"U"VæFÖVçFÅ

238. UPMSPP ÉK" M" *" @'YM"y> 2027 "TM G'IA, •y8 •Y"U'Ú'6/A%G%G%. ÉyÚB3000@)•TM\$ "US" "(y-@
[English]: For UPMSPP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"Ü* 'r' '©M' \$"Ü/YM'r' 'Y5"ä-"Ü0"é:R 5"ü")M'¢ ò ÷ F—(a) 2Y?Y2V6PantodystrainingÉpith Ò vVöÖWG ical meaning of zeroes, relation between zeroes and coefficients) "Y>Y"é"Ü"Y%>Y8M"ö"Y/Y"Option B

Q.239. UPMSP EK" M" * @ YM"y> 2027 "™GIA, y8·Y"U"/6r%G% EypB8B@zr%8 us (y y @ [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?"Y2"Ü* 'r' '©M" \$"Ü/YM"r 'Y5"-ä-"Ü0"é."R 5"ù")M"¢ ð ÷ F—09 2"—=ZV6F.Namonditabed Equations in two Variables (ik '©0"1 "Y>"G "H'i Algebraic methods) "Y> "Y "é"/"U/"Y%">"Y8"IM"ð "Y (/ Option B) Invalid or error message

240. UPMSP É'K " M' *" @'YM"y> 2027 "™G'IA, •y8•Y""U'@BQ'G'™E'N?E•R•™T™•™S™"L\$"™"™"™
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic: Mathematical Principles of Part of Linear Equations in

[illegible]

241. UPMSPP 'ÉK' M' *" @'YM'y> 2027 "™G'IA, •y8 •Y'™"Ú'íé'™%G' % 'É'y?B'8 "™ @ •™\$Y\$ " "(y-@

[English]: For UPMSPP Board Examination 2027, choose the correct answer for this topic.

[illegible]

242. UPMSP 'ÉK' M' *'' @'YM'x> 2027 "TM'G'IA, •y8•Y'''Ü'él' %'G' %' 'É'y?B'8"TM'@•TM'\$"U\$" "(y'~@

[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"U* 'r' '©M" \$"U"/YM"r' 'Y5"â -"U0"é. R 5"ù)"M"¢ ò ÷ F—(a) 2Y2"Y2VEP"riangles (IM" option A Ò 6—Ö—Æ ity criteria, Basic Proportionality Theorem) "Y>Y "é(M"Y%Y%Y>Y8"IM"ò "Y(Y, Option B) Invalid or erroneous statement

Q.243. UPMSP EK M "x" @YM">2027"TMGIA,•y8•Y™U/e/%oG%oE'y?B8™@•™\$™Us\$ "(y-@

A) "Y?Y2'Ü* ' ' ' %o."y %oG 'YK•, ("™@•" ò ÷ F-ôâ B' æðæR öb B) WGPY2'Ü* ' b' 6ö÷ ÷ F-æ FR vVöÖWG y ("%?" M'iG'>•) 'ÉM'ù>é?I? - Distance formula

Coordinate Geometry ('%o?" M'iG"i>•) 'ÉM"ù>é'?I? - Distance formula, section

Q.244. UPMSP 'ÉK" M' *" @'YM"y> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM@ •TM\$"Ù\$" " ('y -@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y"?Y2U" *'R'-çG&ôGV7F-ôâ F0 Trigonometry ('IM'?YK9."üB)yA Y2TÜ/bTrigG&ôGV7F-reâdS0r trigonometry('IM'?YKCM)"\$>-@P'RZQ principle of introduction to Trigonometry ('IM'?YK9."\$>-ô â""?Sân Tainbvalnesit'yvafis'B'Sin yad'Myayes)M'e"/X// Option B) Invalid or error
C) Y"?Y2U"- @M "\$ U/YM" Y5a-U0e.R5ü")Mç ô-F-ôâ ZY-zkveEfvah distractioñ ßunföñ "@_o_F-ôâ Bz æder öb FTW6P Introduction to Trigonometry ('IM'?YK9."\$>-ô â""?©/- Trigonometric ratios

Q.245. UPMSP 'ÉK" M' ** @'YM"> 2027 "TM G'IA, -y8-Y"Ù'Vel1%G "%o 'É'y?'B 8"TM @ •TM\$"Ù\$" "(y-@
[English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

Q.246. UPMSP 'ÉK' M' ** @'YM"> 2027 "™G'IA, •y8•Y™Ù'É'st66G's% É'y?B8™@•™\$™Ù\$ " ("y-@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?"Y2"Ü* 'r' '©M' \$"Ü/YM"r 'Y5'a -"Ü0"é. R 5"ù)"M"ç ò ÷ F—&6/EW2 %5%9\$Ü\$ - Tangents to a circle, theorems on external tangents)
B) "Y?"Y2"Ü* 'r' '©M' \$"Ü/YM"r 'Y5'a -"Ü0"é. R 5"ù)"M"ç ò ÷ F—&6/EW2 %5%9\$Ü\$ - Tangents to a circle, theorems on external tangents)
C) "Y?"Y2"Ü* 'R' 6—&6/EW2 %5%9\$Ü\$ - Tangents to a circle, theorems on external tangents)
D) "Y?"Y2"Ü* 'R' 6—&6/EW2 %5%9\$Ü\$ - Tangents to a circle, theorems on external tangents)

Q.247. UPSPSP EK* M " @ YM y> 2027 TM G IA, y8 Y"U" e/ %G %E y? B 8 TM @ • TM \$ "Ü\$ " ("y -@
[English]: For UPSPSP Board Examination 2027, choose the correct answer for this topic.

C) "Y?Y2"Ü* 'r' '©M" \$"Ü/YM"r 'Y5â -"Ü0é.R 5ü")M"ç ò ÷ F—ôä B' æðæR öb B|W6PY2"Ü* 'b' &V 2&V/E FVB Fò 6—&6EW2 %05'9\$"Ü\$1 %0G %00 'É 'y?'E

Q.248. UPSP 'ÉK" M' ** @'YM"> 2027 "TM'G'IA, •y8•Y"Ü"• of Areas Related to Circles (YC IM IK "8'r 8•y)•
 [English]: For UPSP Board Examination 2027, choose the correct answer for this topic.

[illegible]

Q.249. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM'G'IA, y8•Y"Ù'/'é/ %oG %o 'É 'y?'B 8"TM @ •TM\$"Ù\$" " ('y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

A) "Y?Y2"U* 'b' 7F F—7F—72 %ø"é 'IM'ù?Y@ - Mean, Median, Mode of grouped data) —7F—72ø(8% YM'ù>Y@C (Mean, Median, Mode of grouped data)

Q.250. UPMSP 'ÉK" M' *" @'YM"> 2027 "TM'G'IA, •y8•Y"Ù'/'é/ "‰G "‰ 'É 'y?'B 8"TM@ •TM\$"Ù\$ " ("y -@
 [English]: For UPMSP Board Examination 2027, choose the correct answer for this topic.

[illegible]